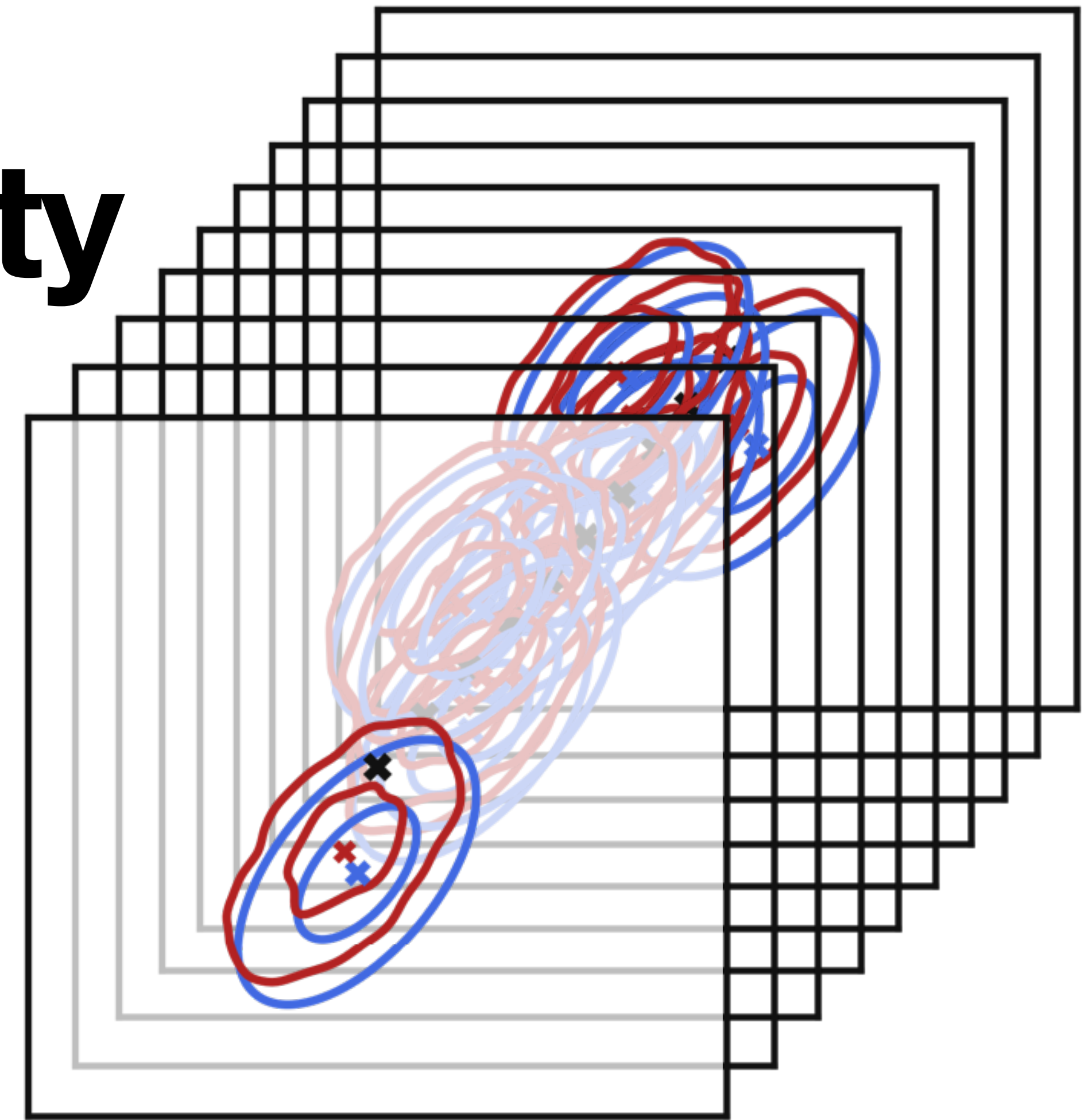


Bayesian and Frequentist Probability

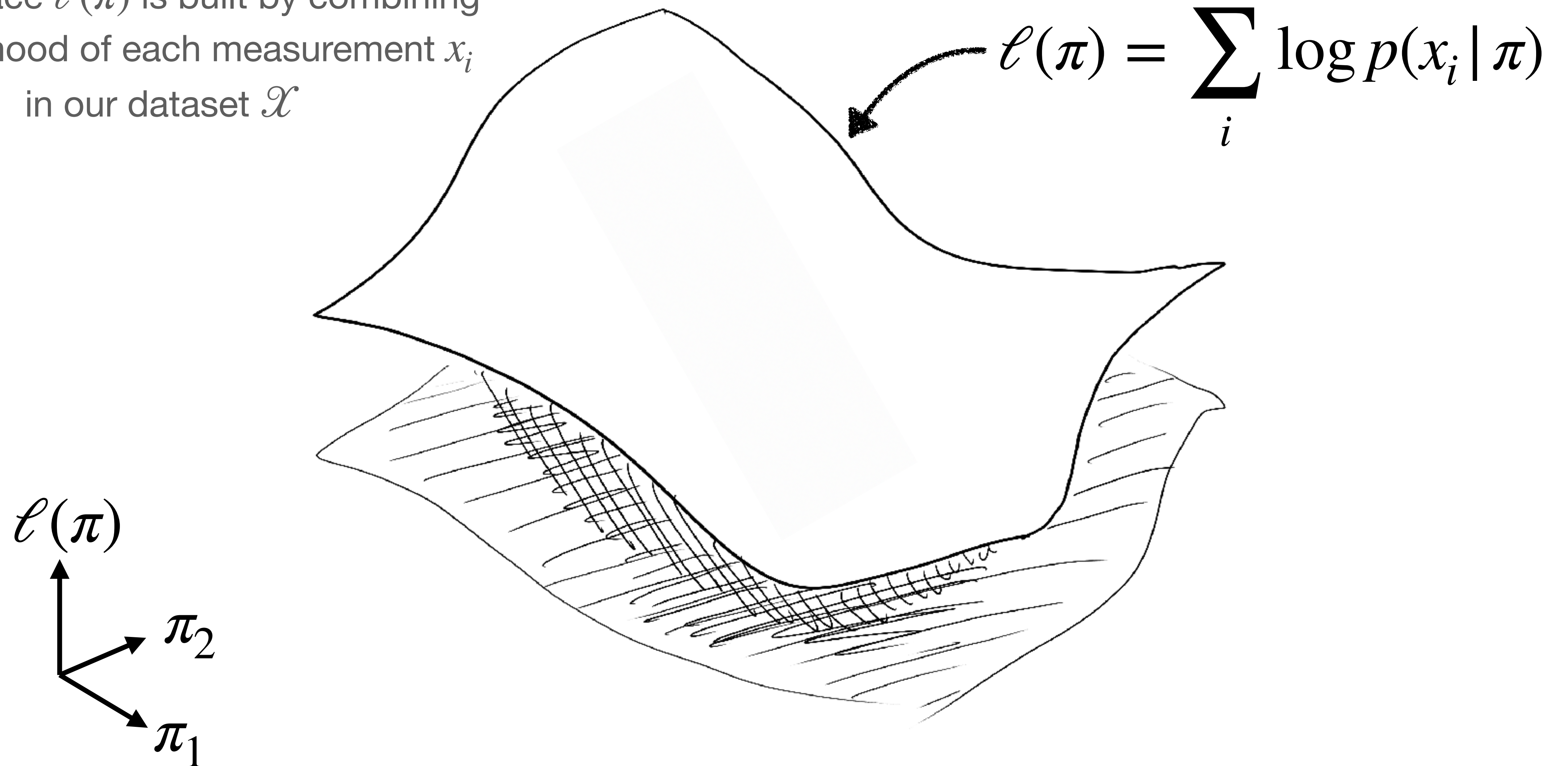
Jed Homer LMU/ODSL



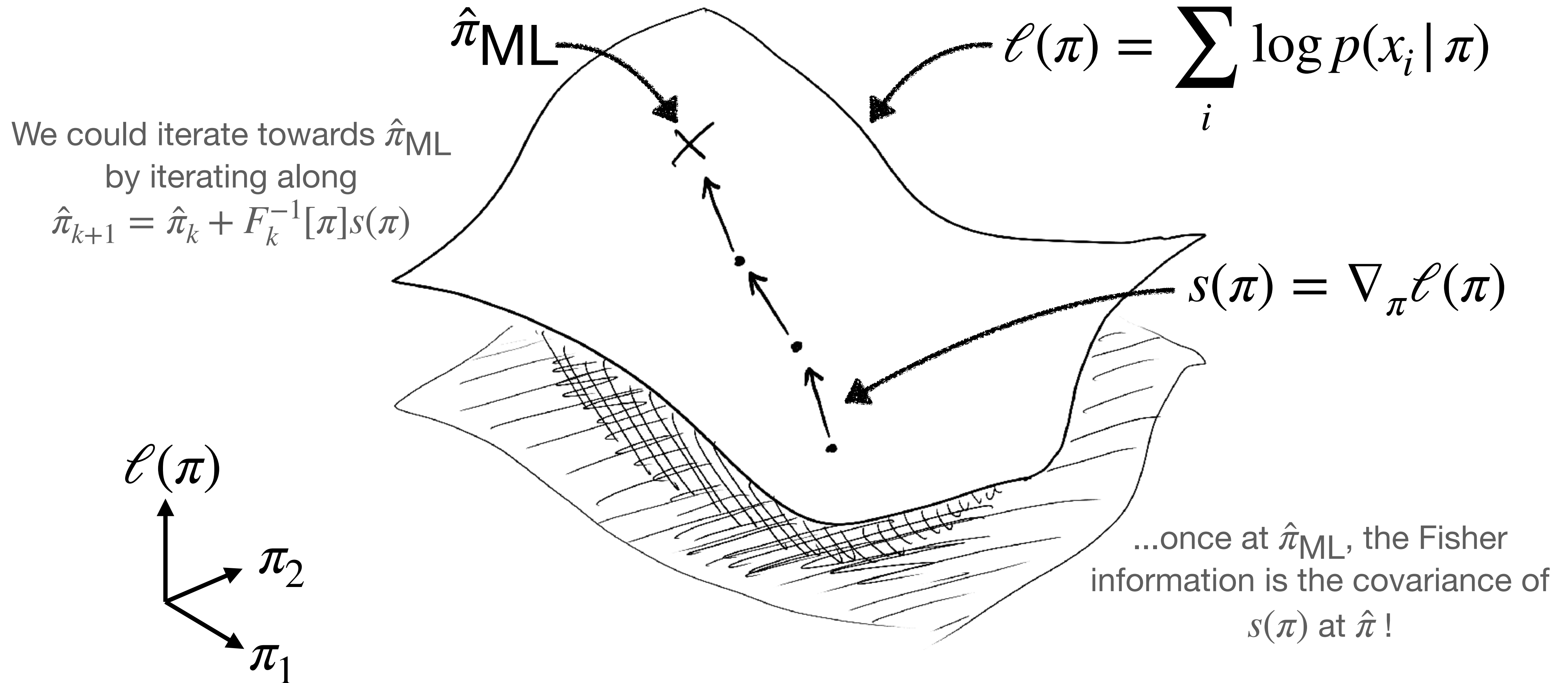
$$\pi \quad p(\hat{\pi} | \pi) \quad p(\pi | \hat{x})$$

Previous lecture: Fisher information

This surface $\ell(\pi)$ is built by combining the likelihood of each measurement x_i in our dataset $\mathcal{X}$



Previous lecture: Fisher information

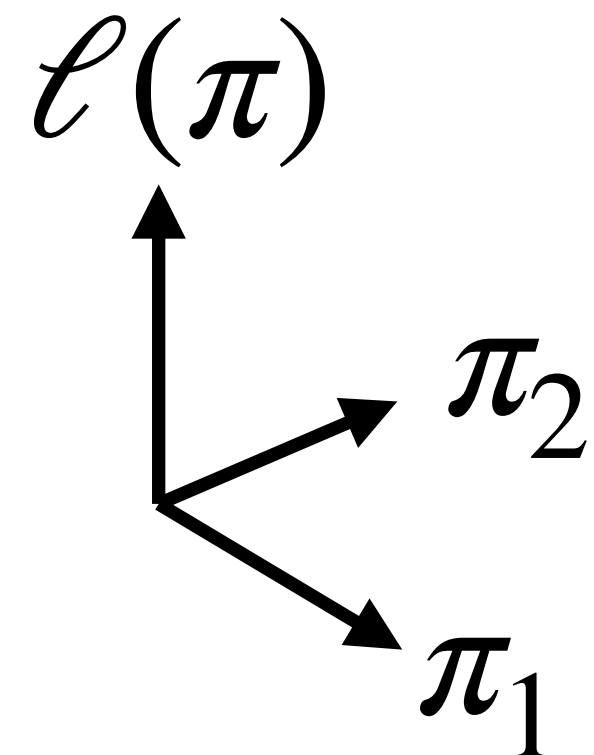


Previous lecture: Fisher information

$$F[\pi] = \text{Cov}_{\mathcal{X}}[s(\pi)s(\pi)^T]$$

Fisher information is the covariance of this gradient vector $s(\pi)$!

Remember that $s(\pi) = 0$ at $\hat{\pi}_{\text{ML}}$!



$$\ell(\pi) = \sum_i \log p(x_i | \pi)$$

Previous lecture: Fisher information for $\hat{\mu}$

- Revisit example of estimating μ using a 2D dataset $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$
- Calculate $F(\mu) = \mathbb{E}_x[\mathbf{s}(\mu)\mathbf{s}(\mu)^T]$ by differentiating $\mathcal{G}[\mathbf{x} | \boldsymbol{\mu}(\mu), \Sigma]$ w.r.t. μ

$$\boldsymbol{\mu}(\mu) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \mu$$

$$\Rightarrow \nabla_{\mu} \boldsymbol{\mu} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{aligned} \Rightarrow F(\mu) &= \nabla_{\mu} \boldsymbol{\mu} \Sigma^{-1} (\nabla_{\mu} \boldsymbol{\mu})^T \\ &= \text{Trace}[\Sigma^{-1} \mathbf{1}^T \mathbf{1}] = n \Sigma^{-1} \end{aligned}$$

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$$\Rightarrow F(\mu) = \nabla_{\mu} \boldsymbol{\mu} \Sigma^{-1} (\nabla_{\mu} \boldsymbol{\mu})^T$$

$$= \text{Trace}[\Sigma^{-1} \mathbf{1}^T \mathbf{1}] = n \Sigma^{-1} \Rightarrow \text{Cov}(\hat{\mu} | \mu) = F^{-1}(\mu)$$

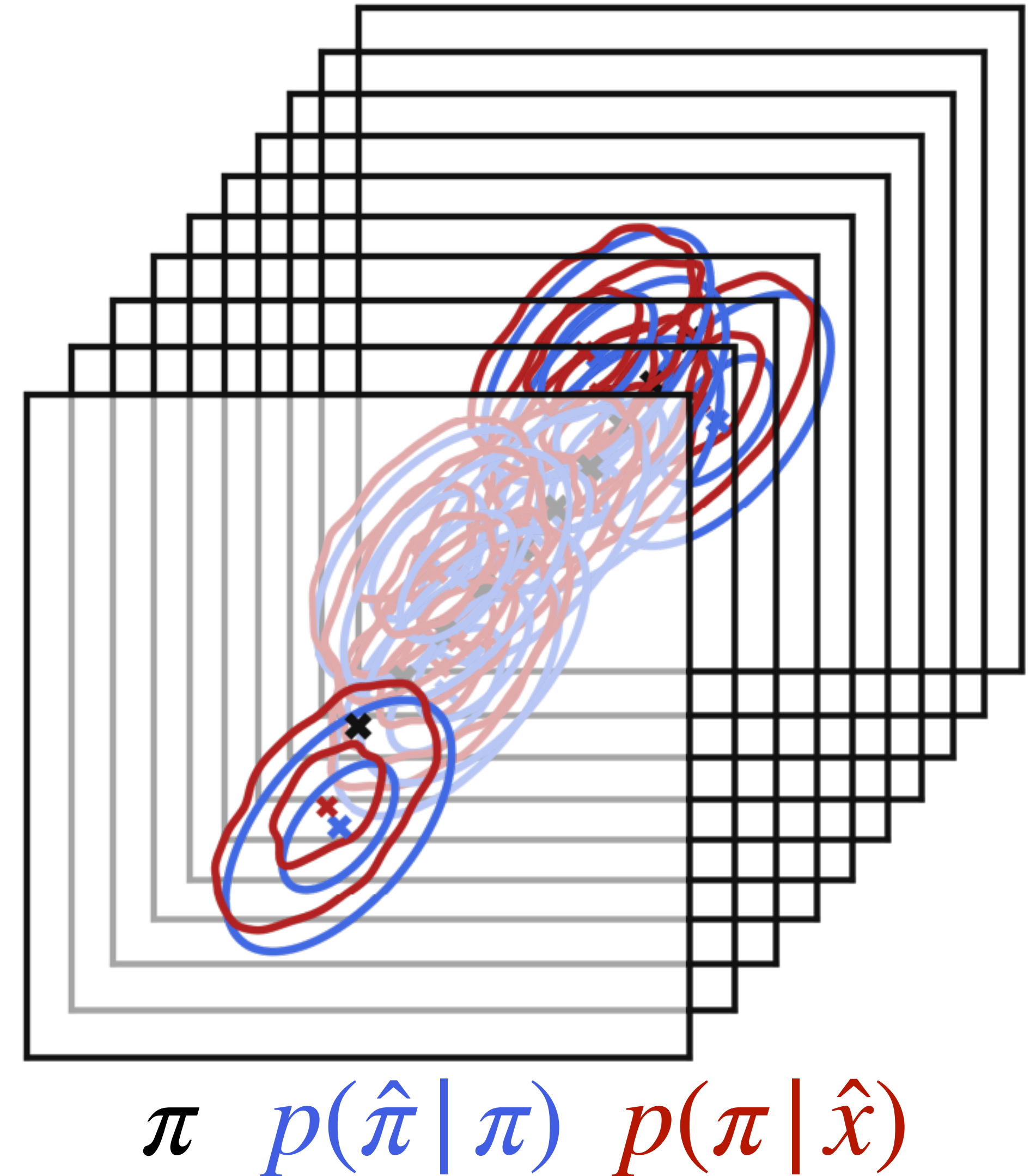
$$p(\mathbf{x} | \mu) = \mathcal{G}[\mathbf{x} | \boldsymbol{\mu}(\mu), \Sigma]$$

$$\Rightarrow p(\hat{\mu} | \mu) = \mathcal{G}\left[\hat{\mu} | \boldsymbol{\mu}(\mu), \frac{\Sigma}{n}\right]$$

$$\Rightarrow \text{Cov}(\hat{\mu} | \mu) = \frac{\Sigma}{n}$$

Synopsis

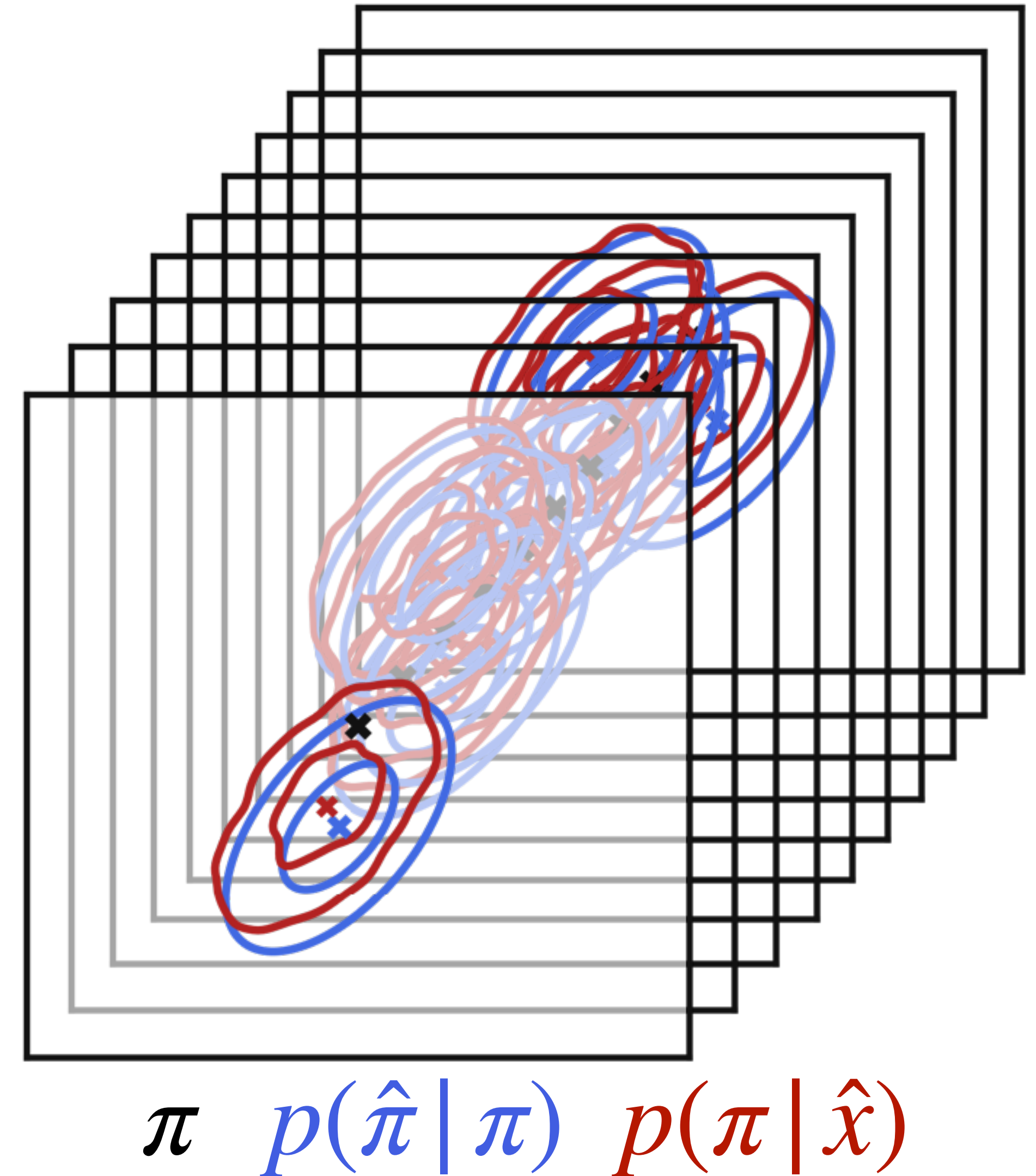
- When comparing theory $x[\pi]$ to data $\hat{x}$, we assume an interpretation of the estimate $\hat{\pi}$ and its 'uncertainty'.



A set of repeated experiments to measure π with the estimator $\hat{\pi}$. Confidence intervals (blue) and credible regions (red) are also shown.

Synopsis

- When comparing theory $x[\pi]$ to data $\hat{x}$, we assume an interpretation of the estimate $\hat{\pi}$ and its 'uncertainty'.
- What are the main ways of interpreting this statement?
- Where do they differ?
- What are the consequences?

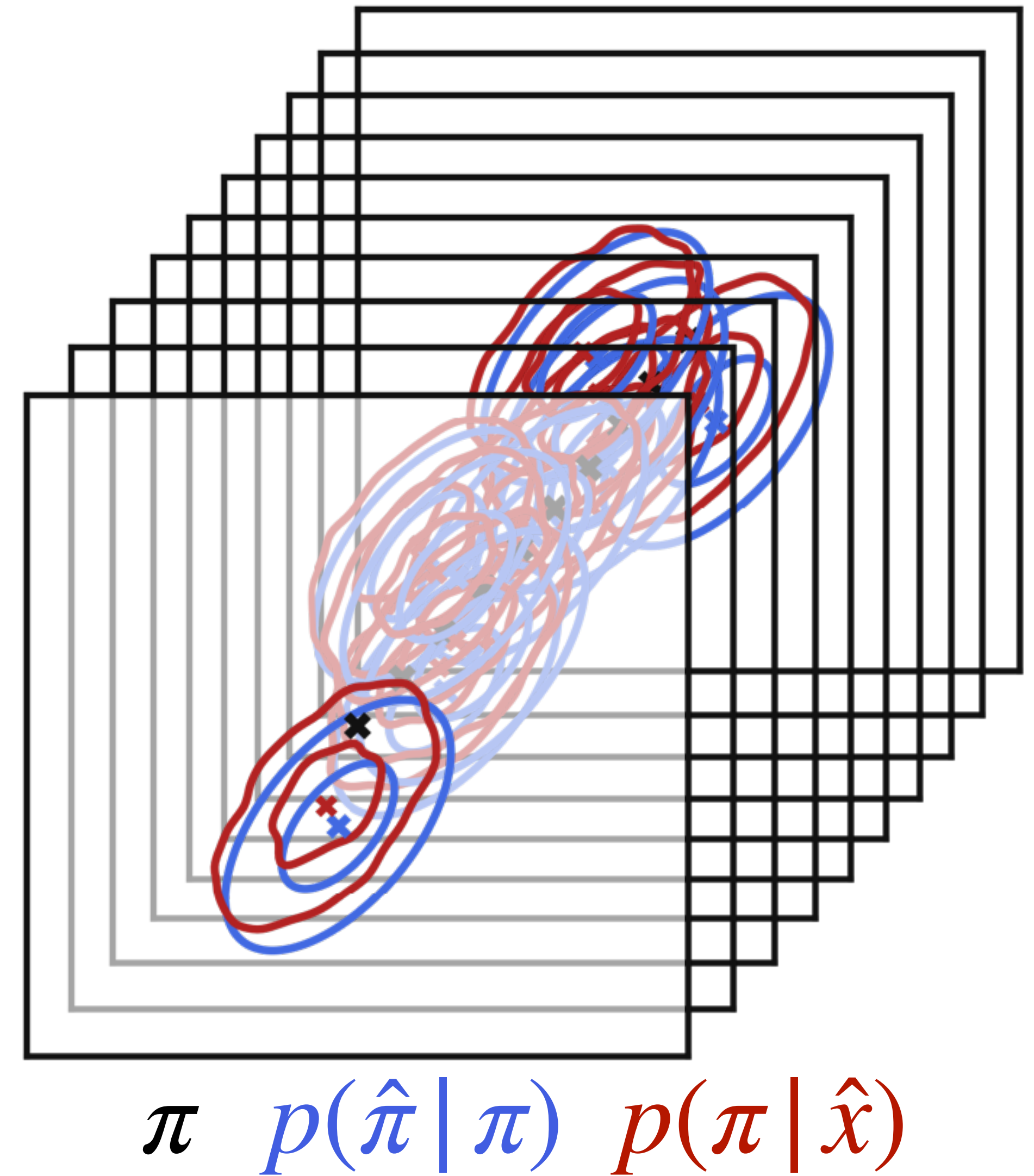


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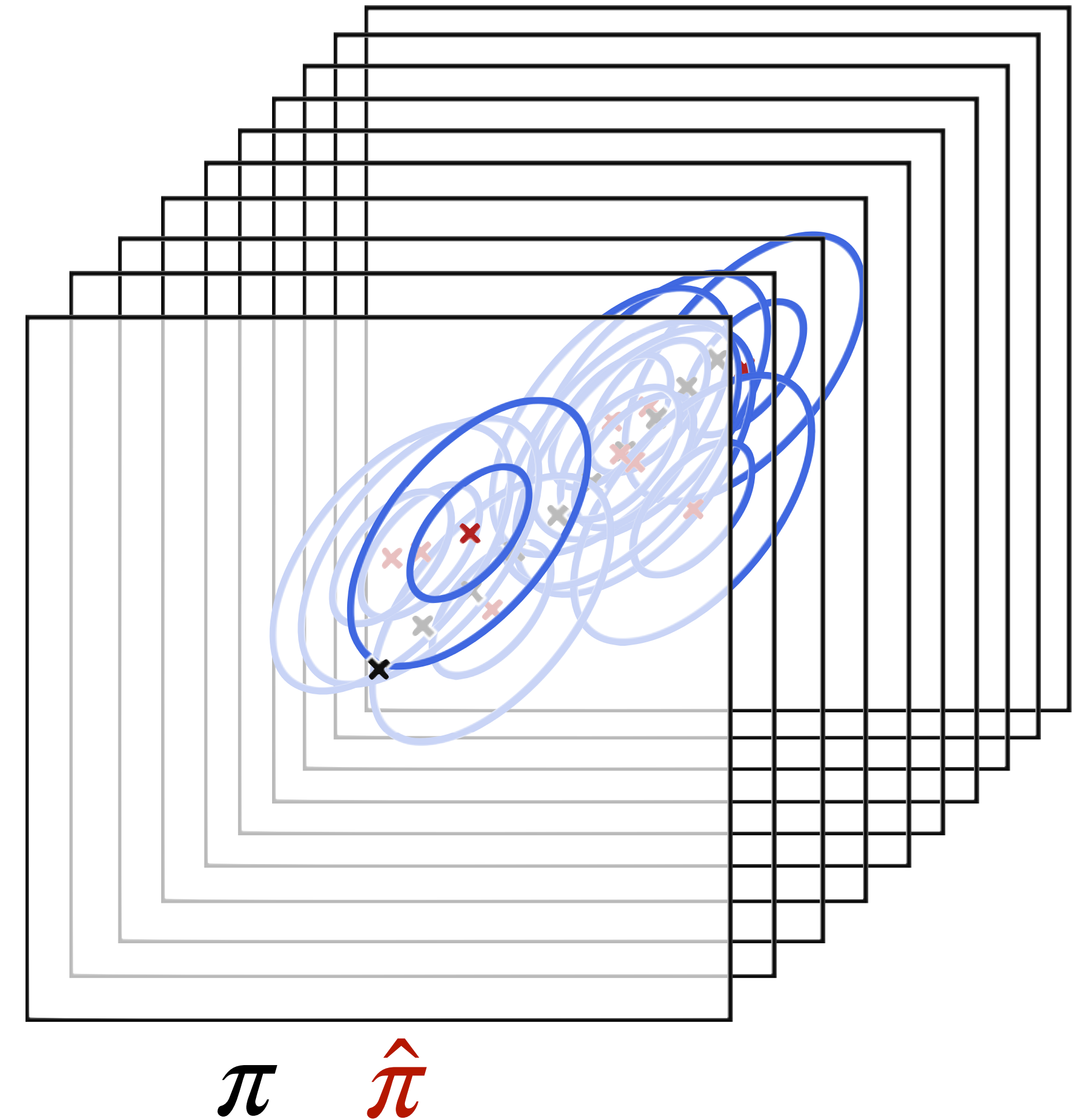
Today: compare the interpretations of Bayesian and Frequentist probability in a simple experiment.



A set of repeated experiments to measure π with the estimator $\hat{\pi}$. Confidence intervals (blue) and credible regions (red) are also shown.

Frequentist probability

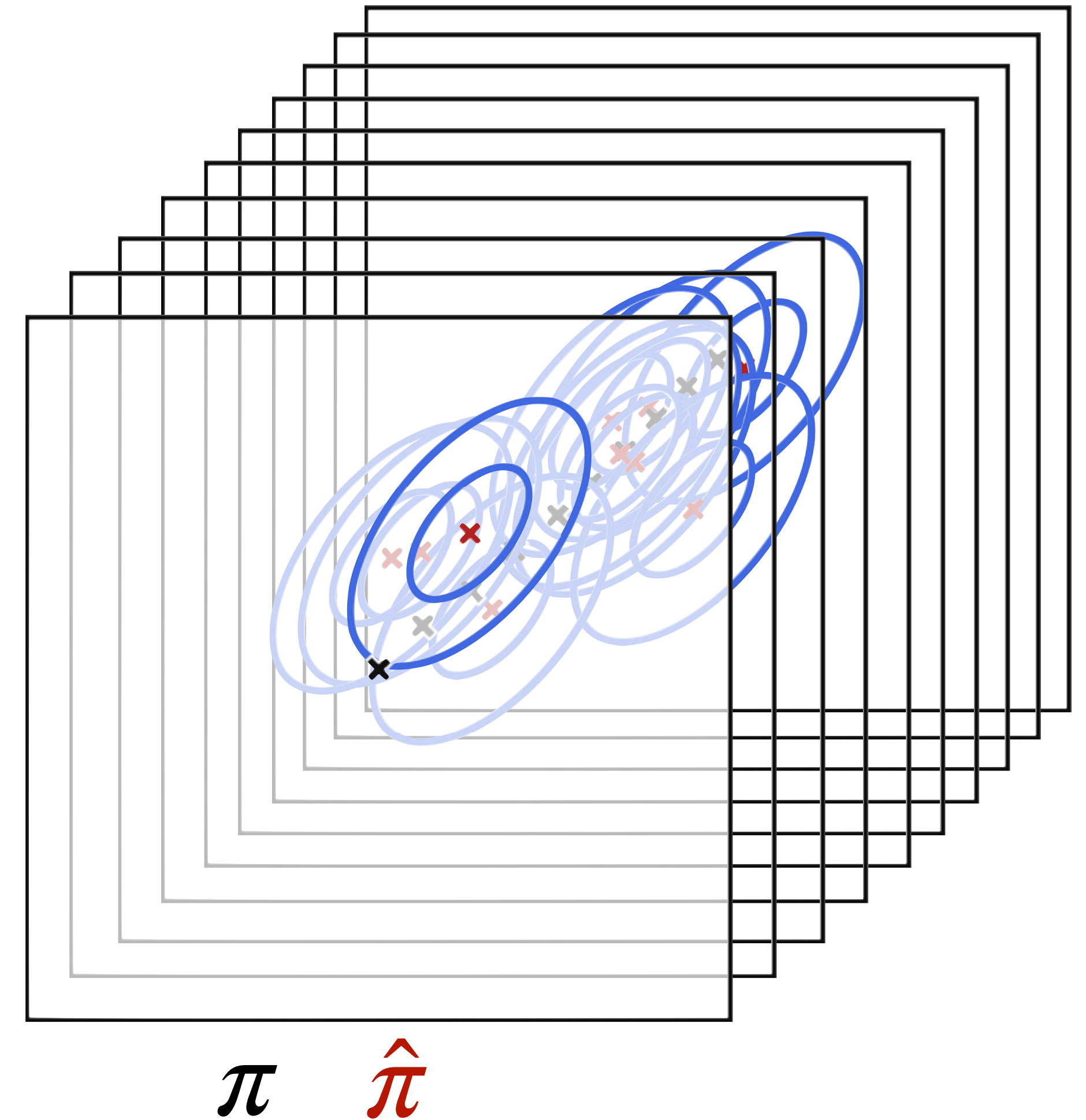
- **Frequentism** interprets probability as the *relative frequency* of each possible outcome.
- The outcomes originate from *identical* experiments.
- Probability of an outcome is the *relative frequency*.
- This requires an infinite sequence of these experiments.



A set of repeated experiments to measure π with the estimator $\hat{\pi}$. Confidence intervals are also shown in blue.

Frequentist probability

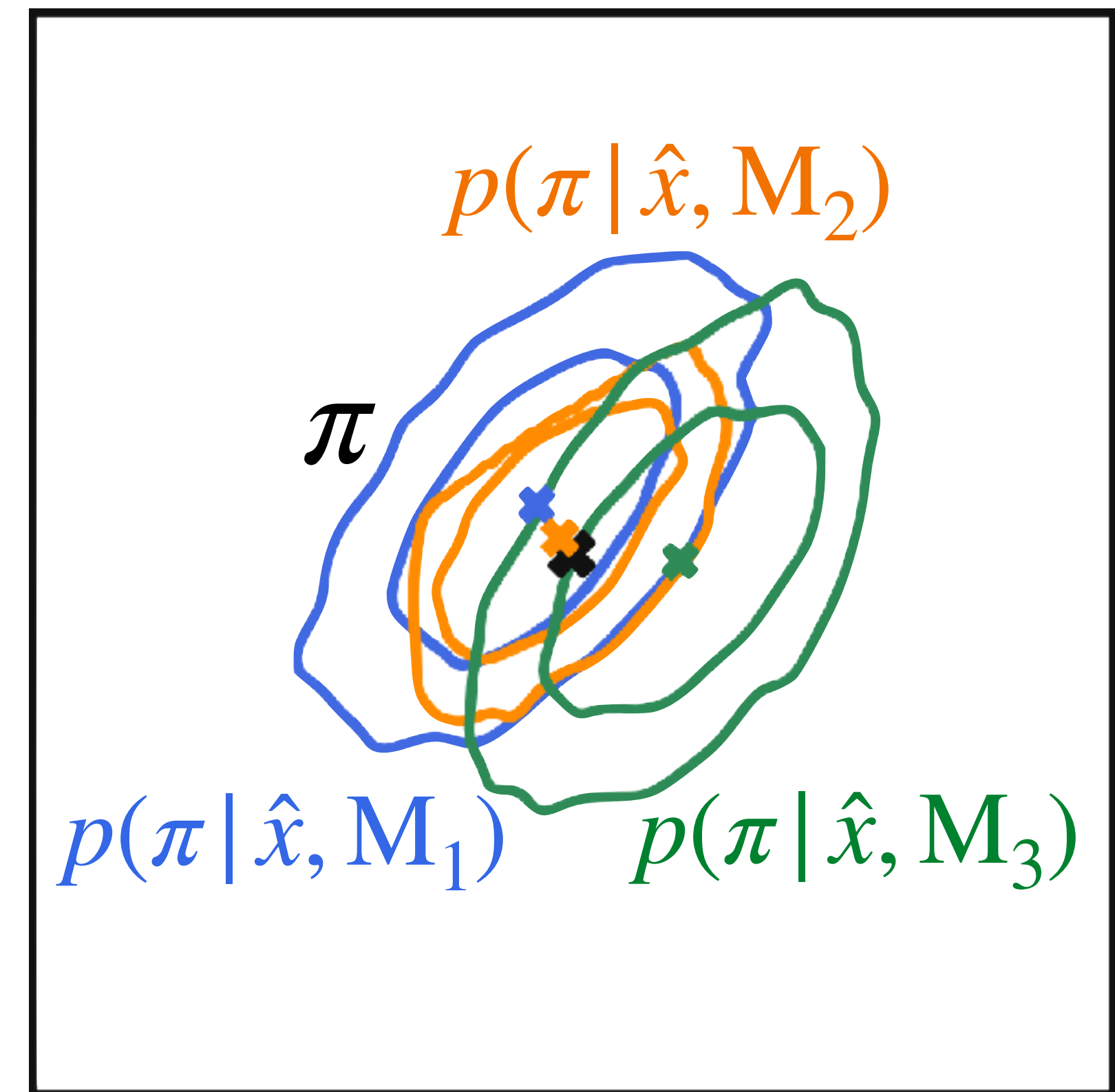
- Physical parameters π have fixed values in this interpretation.
- It makes no sense to vary such parameters, they have no PDF.
- Only an estimate $\hat{\pi}$ does: we are in some identical experiment, and we measured some value.



A set of repeated experiments to measure π with the estimator $\hat{\pi}$. Confidence intervals are also shown in blue.

Bayesian probability

- **Bayesian probability** is a *subjective degree-of-belief* in a statement.
- We do *not* assume there is some true fixed value of what we want to infer.
- Events that have not happened yet can be analysed: we are only ignorant of them.



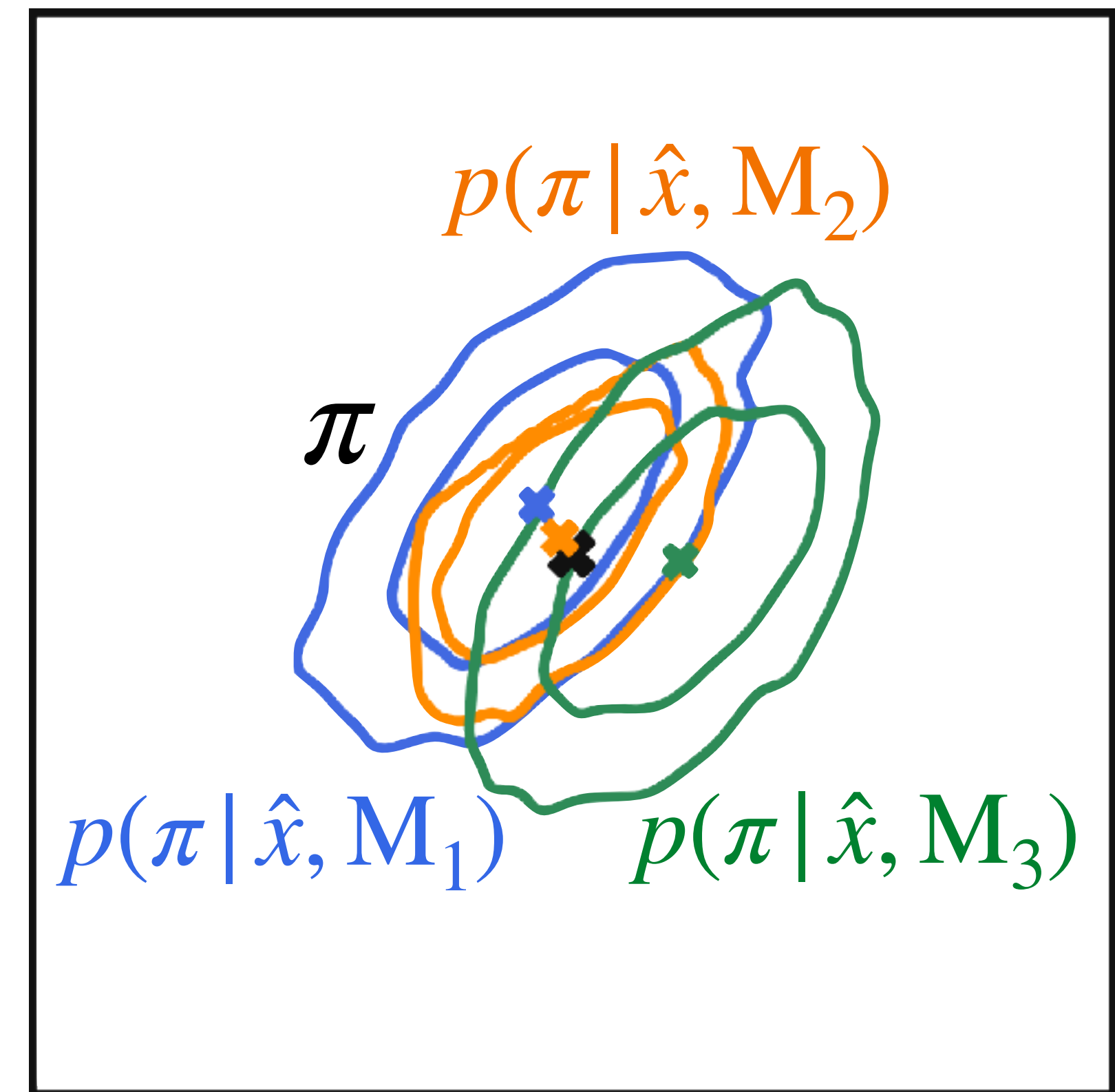
A set of Bayesian analyses with different prior assumptions M_i .

Bayesian probability

- The **Bayesian** summarises their prior knowledge with a prior PDF $p(\pi)$.
- This knowledge is updated in light of data $\hat{x}$, by a likelihood PDF $p(\hat{x} | \pi)$.
- The posterior PDF $p(\pi | \hat{x})$ quantifies their knowledge after measuring the data

$$p(\pi | \hat{x}) \propto p(\hat{x} | \pi)p(\pi)$$

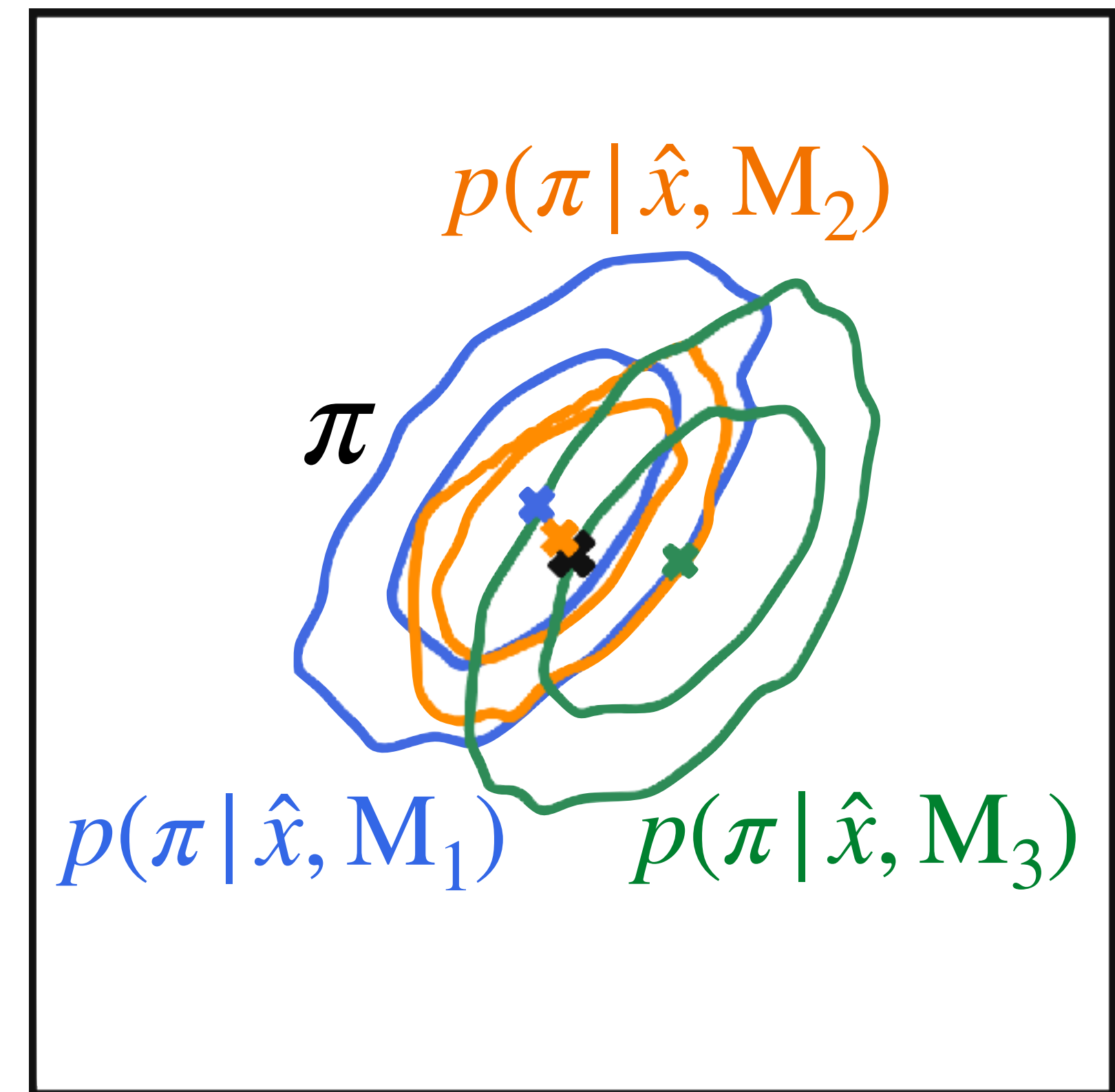
posterior $\propto$ likelihood $\times$ prior



A set of Bayesian analyses with different prior assumptions M_i .

Bayesian probability

- Given the same *prior information* $p(\pi)$, the Bayesian has one PDF to derive: the *posterior* $p(\pi | \hat{x})$.
- Without a huge amount of data, the posterior depends strongly on the *prior*.
- It is still sensible to ask: *what would identical experiments return?*

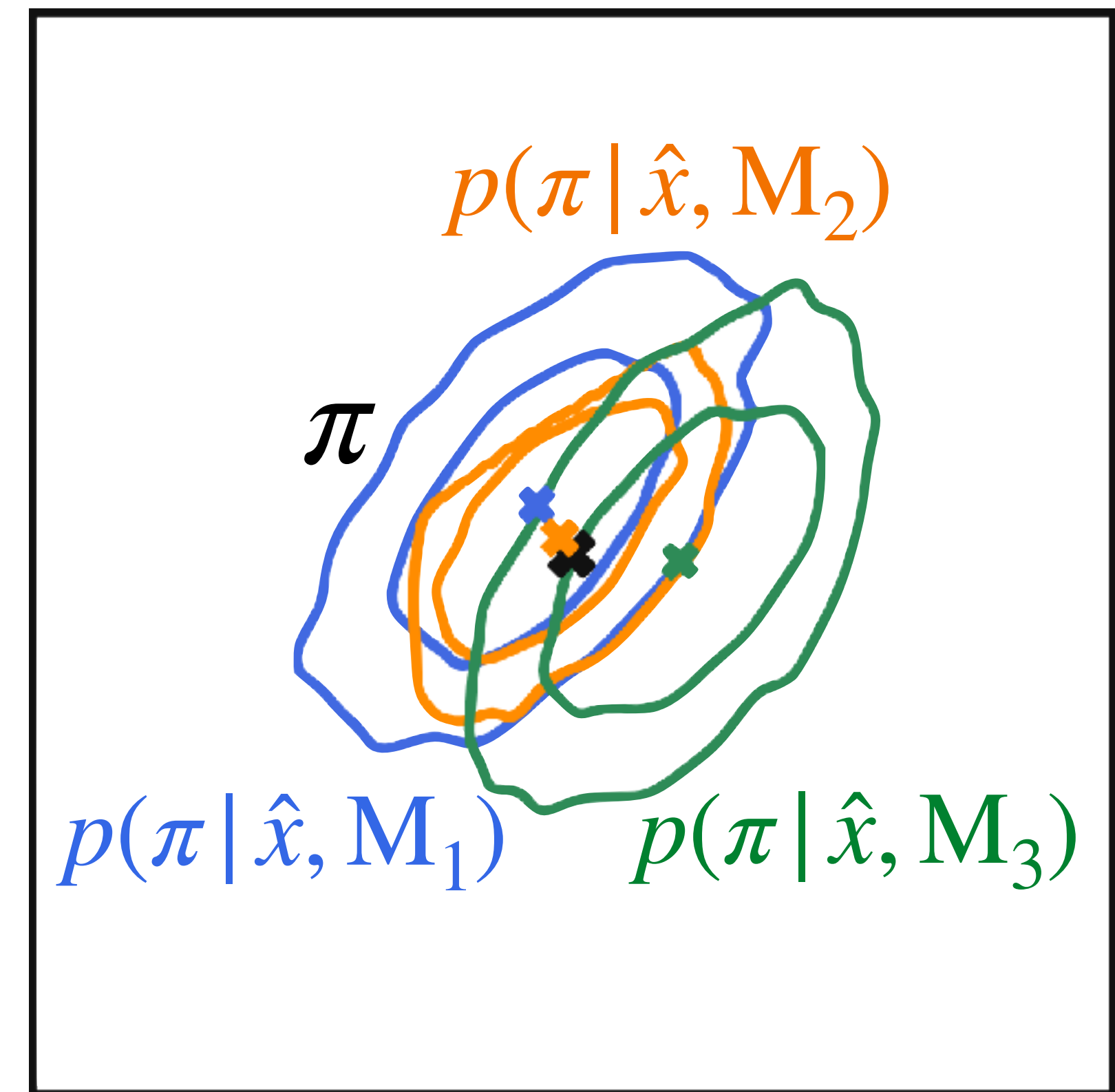


A set of Bayesian analyses with different prior assumptions M_i .

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What does all of this mean to you?
We'll look deeper into this later...



A set of Bayesian analyses with different prior assumptions M_i .

Bayesian probability might not be what you think...

- Let's start with something simple.

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$$\hat{x} \sim p(\hat{x} \mid \pi)$$

Bayesian probability might not be what you think...

- Let's start with something simple.

$$\hat{x} \sim \mathcal{G}[\hat{x} \mid \mu[\pi], \Sigma]$$

Bayesian probability might not be what you think...

- Let's start with something simple.

$$\hat{x} \sim \mathcal{G} \left[\hat{x} \middle| \mu[\pi] = \begin{bmatrix} \pi \\ \pi \end{bmatrix}, \Sigma = \begin{bmatrix} \sigma^2 & 0 \\ 0 & \sigma^2 \end{bmatrix} \right]$$

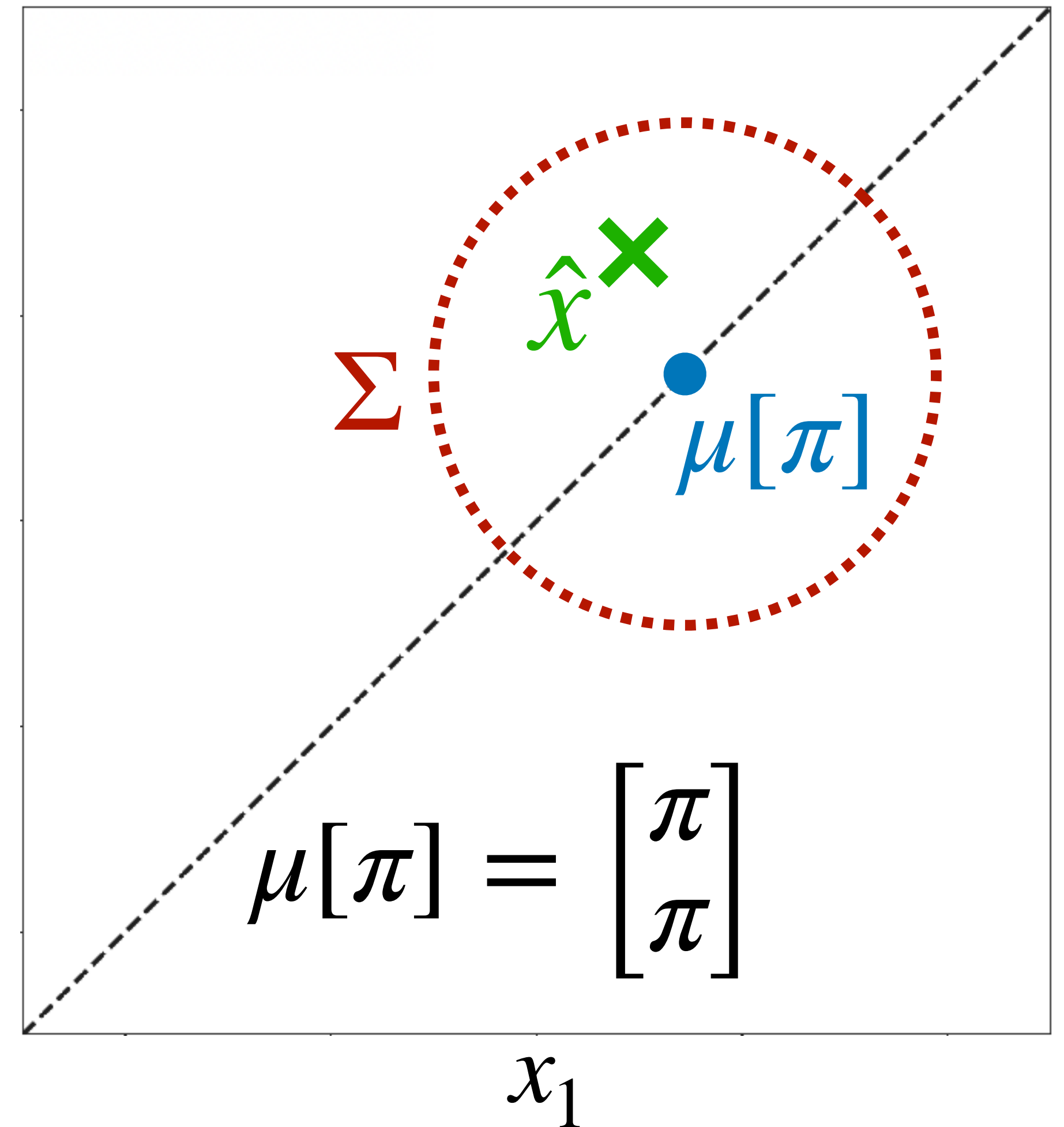
- We know the covariance matrix Σ .
- We'll see what happens when data are generated from different models.

Bayesian probability might not be what you think...

- Let's start with something simple.

$$\hat{x} \sim \mathcal{G} \left[\hat{x} \middle| \mu[\pi] = \begin{bmatrix} \pi \\ \pi \end{bmatrix}, \Sigma = \begin{bmatrix} \sigma^2 & 0 \\ 0 & \sigma^2 \end{bmatrix} \right]_{x_2}$$

- Pick a point $\pi \sim p(\pi)$ and draw $\hat{x}$



Bayesian probability might not be what you think...

- But what do we want from such an analysis?
 - *Roughly speaking*: an estimate $\hat{\pi}$ with a lower and upper bound.

$$\pi \in [\hat{\pi} - \sigma, \hat{\pi} + \sigma]$$

- How do *you* interpret this?

Bayesian probability might not be what you think...

- This is a *Frequentist confidence interval* $\mathcal{C}_{68\%}^{CI}$

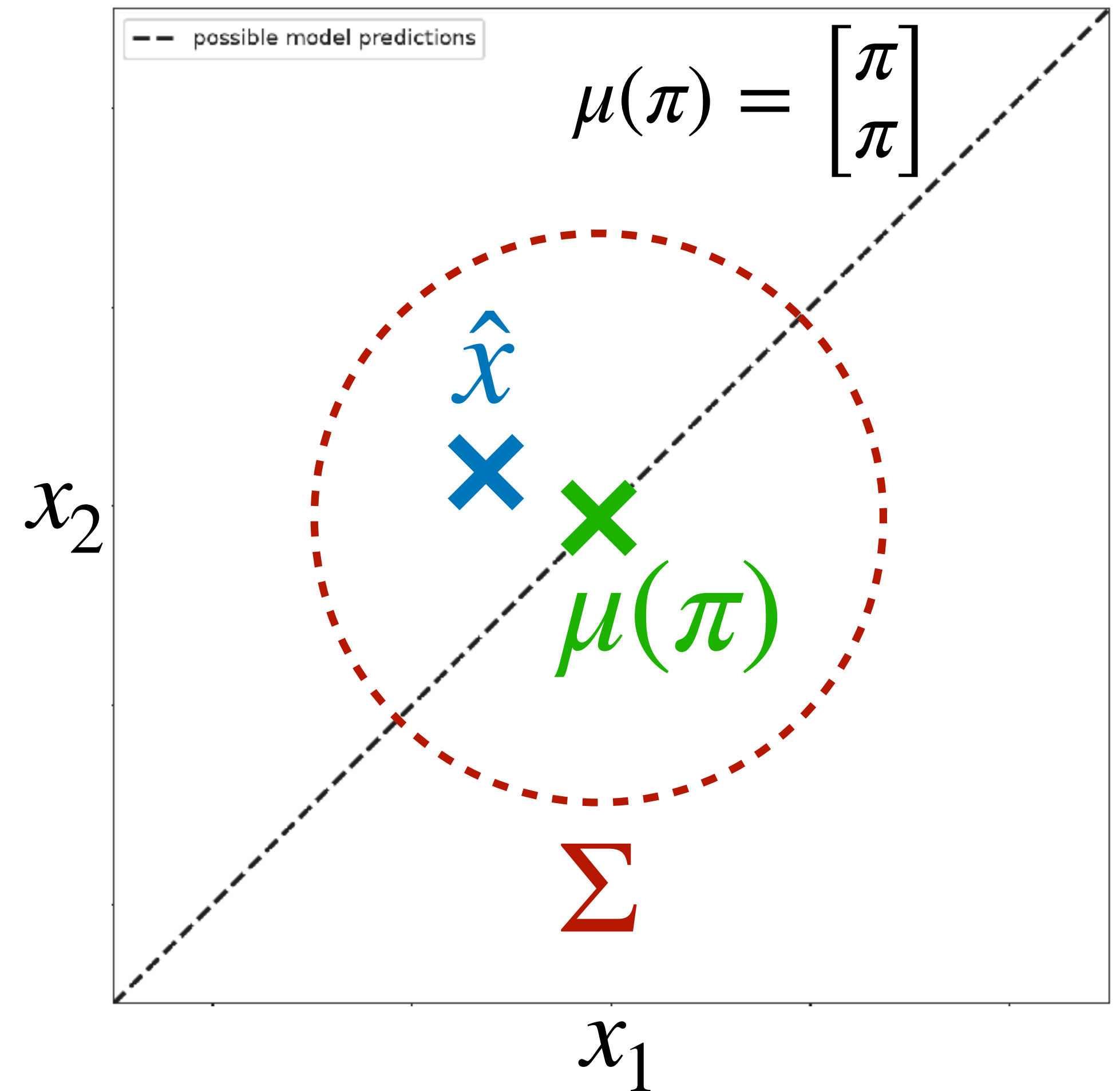
$$\pi \in [\hat{\pi} - \sigma, \hat{\pi} + \sigma]$$

- A Bayesian assigns a *credible region* $\mathcal{C}_{68\%}^{CR}$

$$\int_{\pi \in \Omega} d\pi p(\pi | \hat{\pi}) = 0.68$$

Bayesian probability might not be what you think...

- Let's start with something simple.
 1. We draw data $\hat{x}$ at a point π in parameter space.

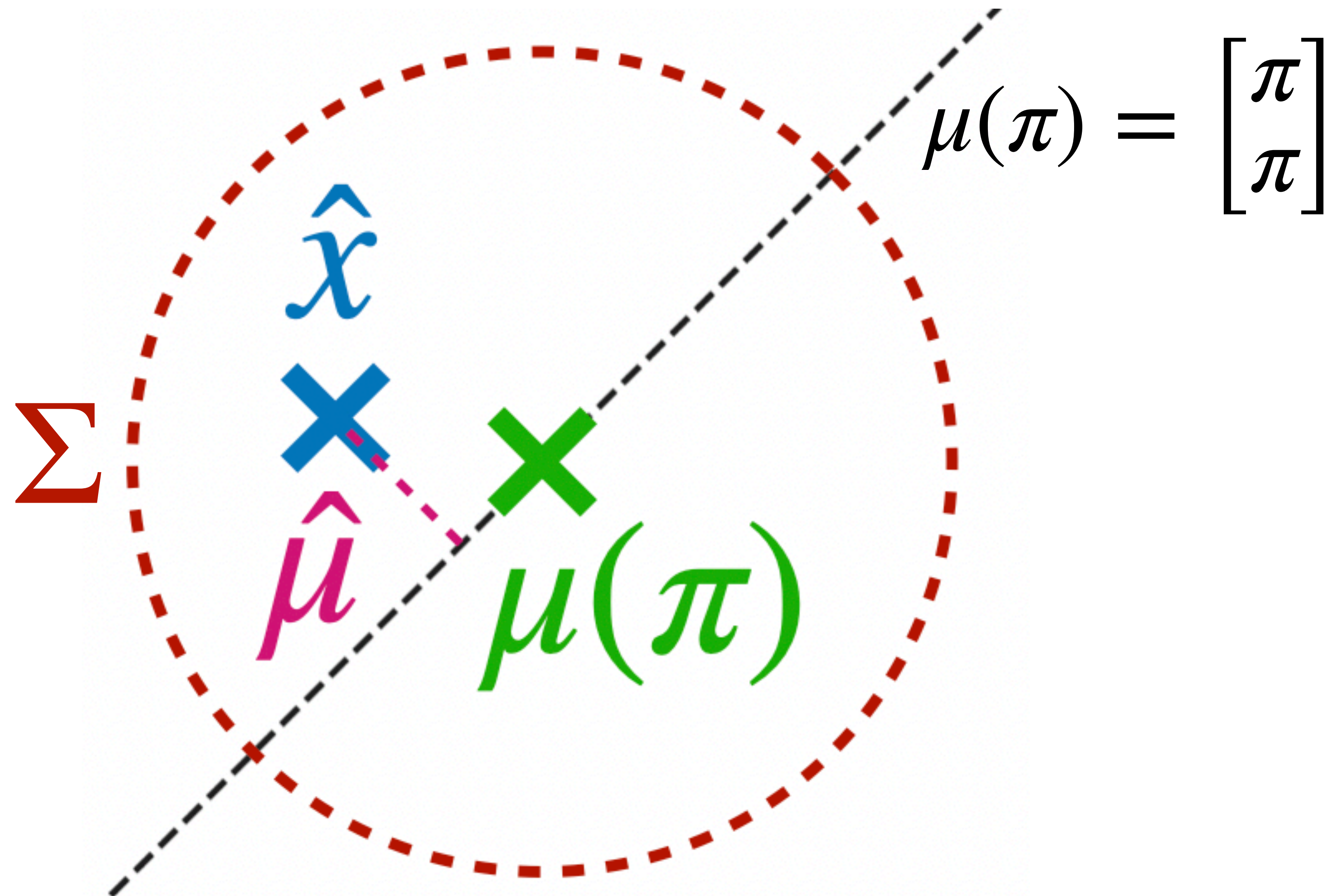


Bayesian probability might not be what you think...

- Let's start with something simple.
 1. We draw data $\hat{x}$ at a point π in parameter space.
 2. Estimate the parameters π with $\hat{\pi}$ (e.g. the MLE).

$$\hat{\pi} = \max_{\pi} \log p(\hat{x} | \pi)$$

What's happening in each experiment?



Bayesian probability might not be what you think...

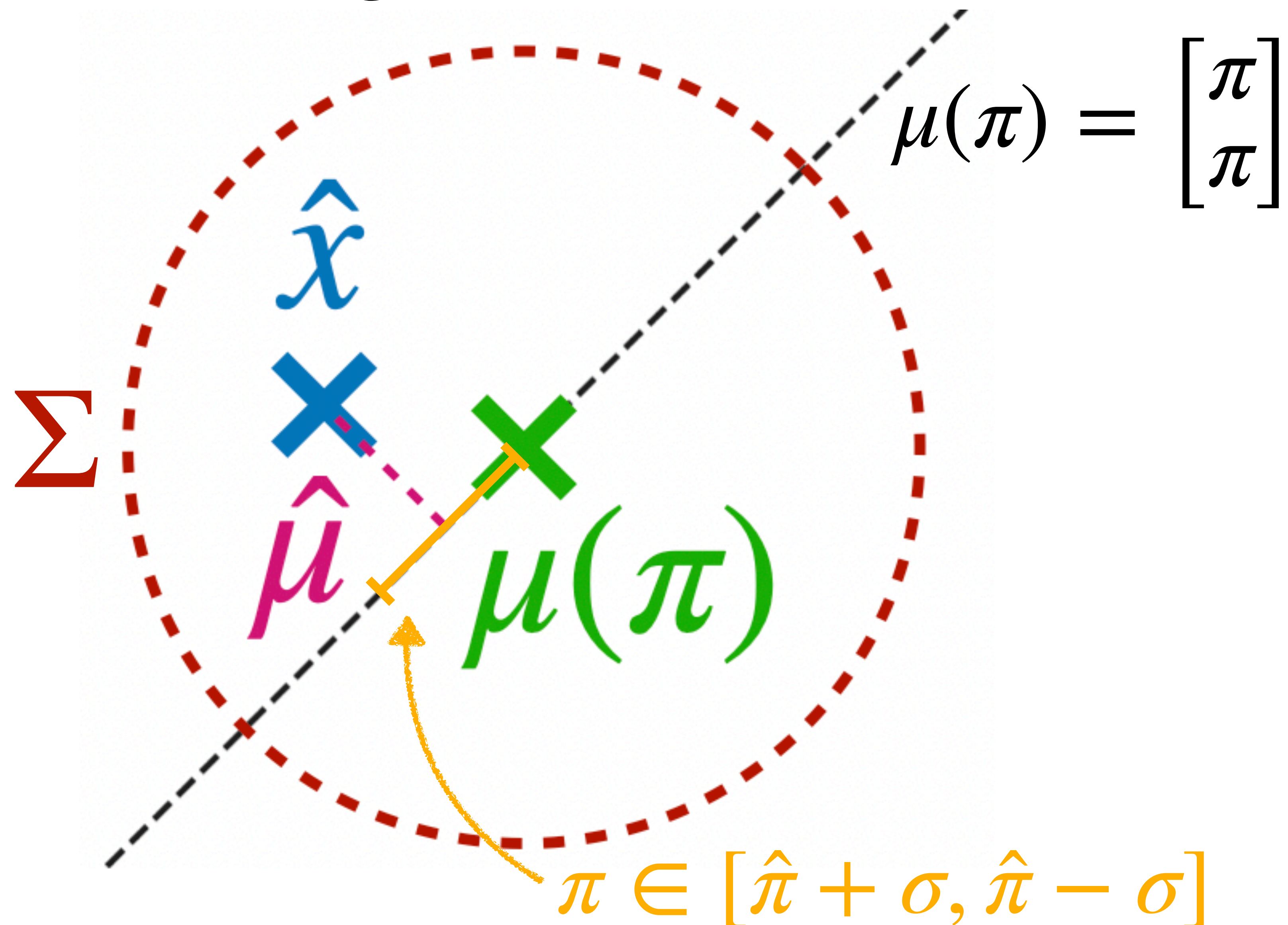
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$$\hat{\pi} = \max_{\pi} \log p(\hat{x} | \pi)$$



$$p(\pi | \hat{x}) \propto p(\hat{x} | \pi)p(\pi)$$

What's happening in each experiment?



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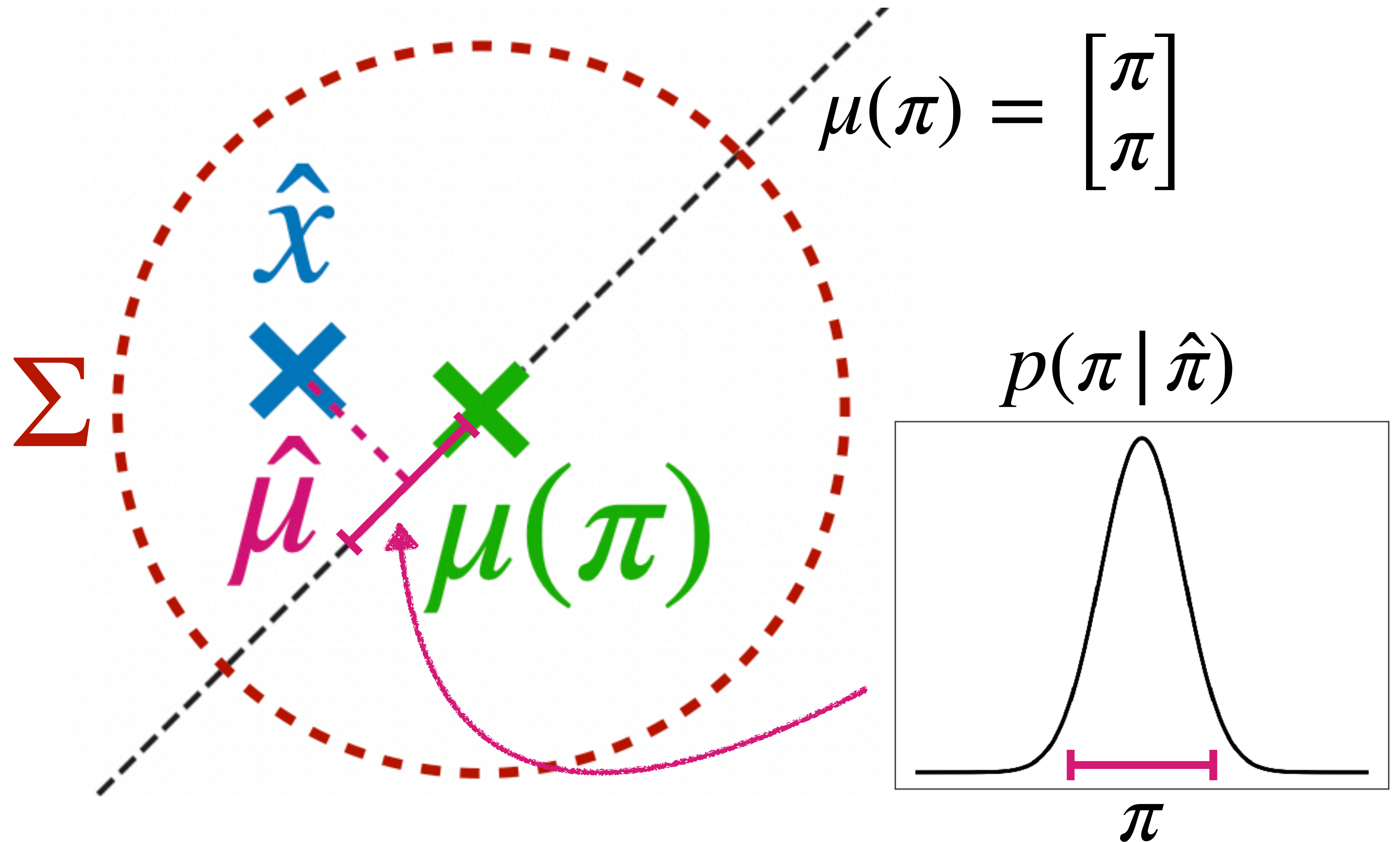
- Let's start with something simple.
 1. We draw data $\hat{x}$ at a point π in parameter space.
 2. Estimate the parameters π with $\hat{\pi}$ (e.g. the MLE).
 3. Calculate a lower/upper interval.
 4. Calculate a posterior....we repeat this many times.

$$\hat{\pi} = \max_{\pi} \log p(\hat{x} | \pi)$$



$$\pi \in [\hat{\pi} - \sigma, \hat{\pi} + \sigma]$$

What's happening in each experiment?

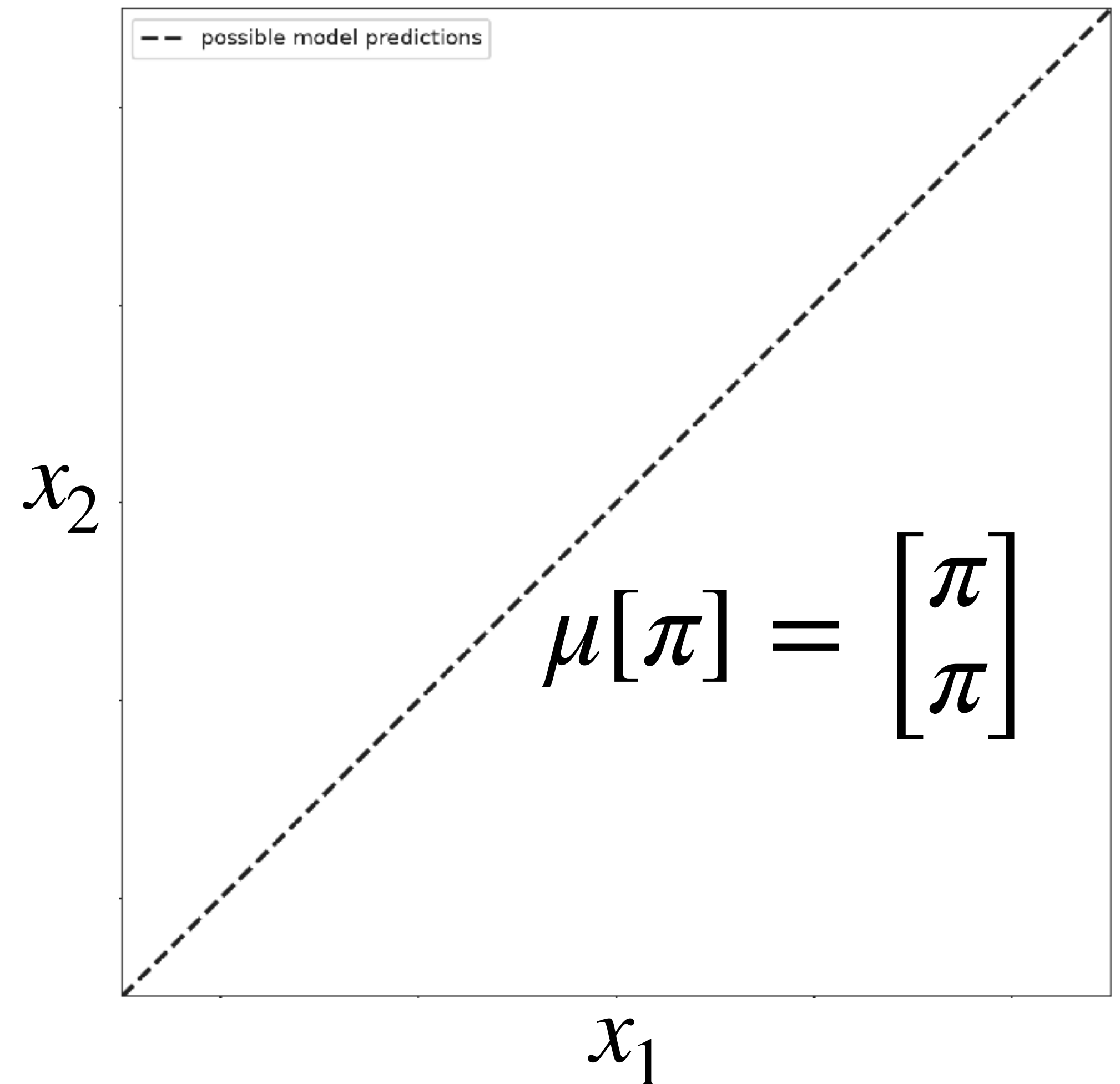


Linear model $x[\pi]$

- This is our simple data generating model

$$\hat{x} \sim \mathcal{G} \left[\hat{x} \left| \begin{bmatrix} \pi \\ \pi \end{bmatrix}, \begin{bmatrix} \sigma^2 & 0 \\ 0 & \sigma^2 \end{bmatrix} \right. \right]$$

- Repeat 10,000 experiments at $\pi = \pi^0$.
- Calculate $\mathcal{C}_{68\%}$: is π_0 **inside** or **outside**?

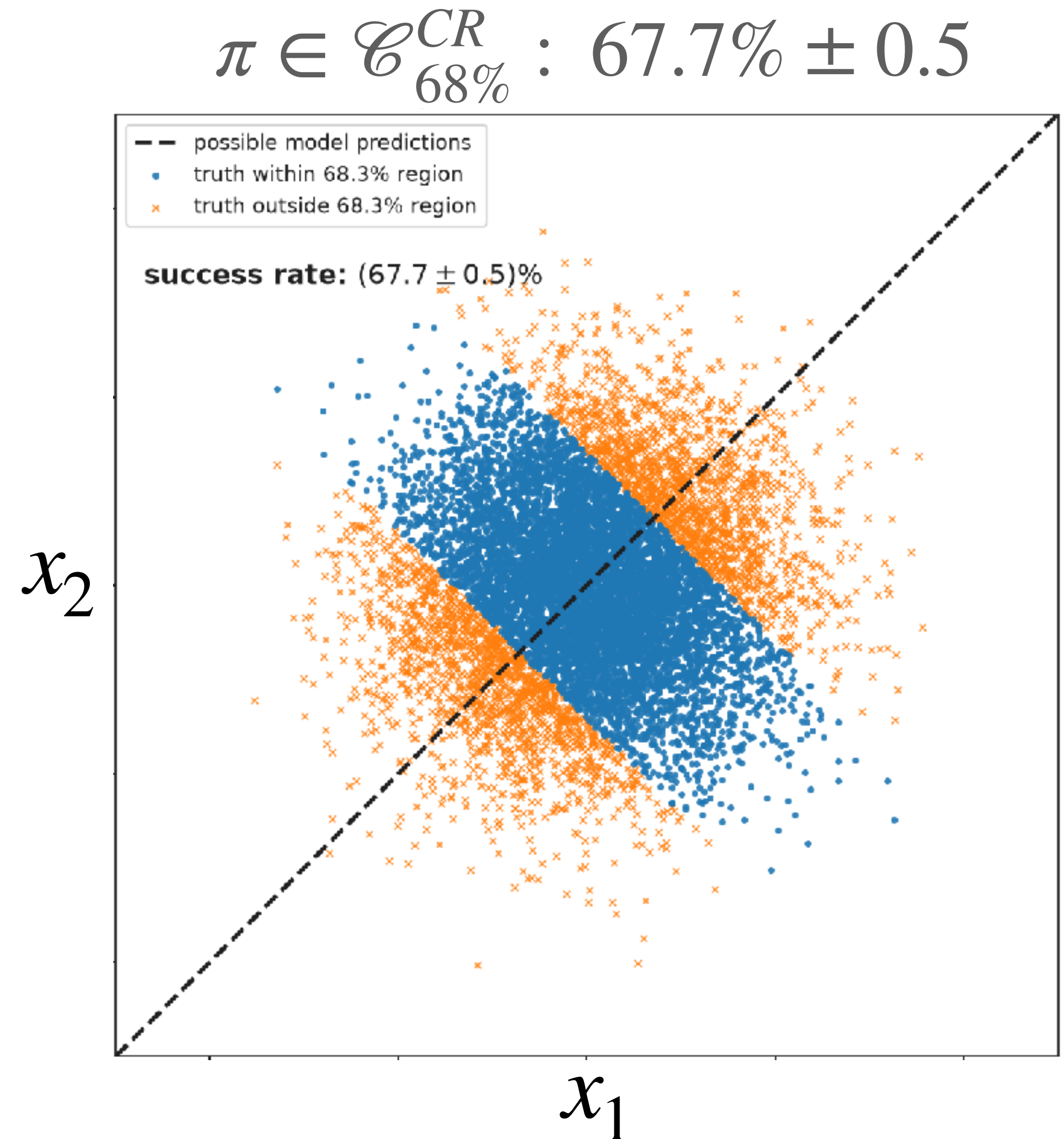


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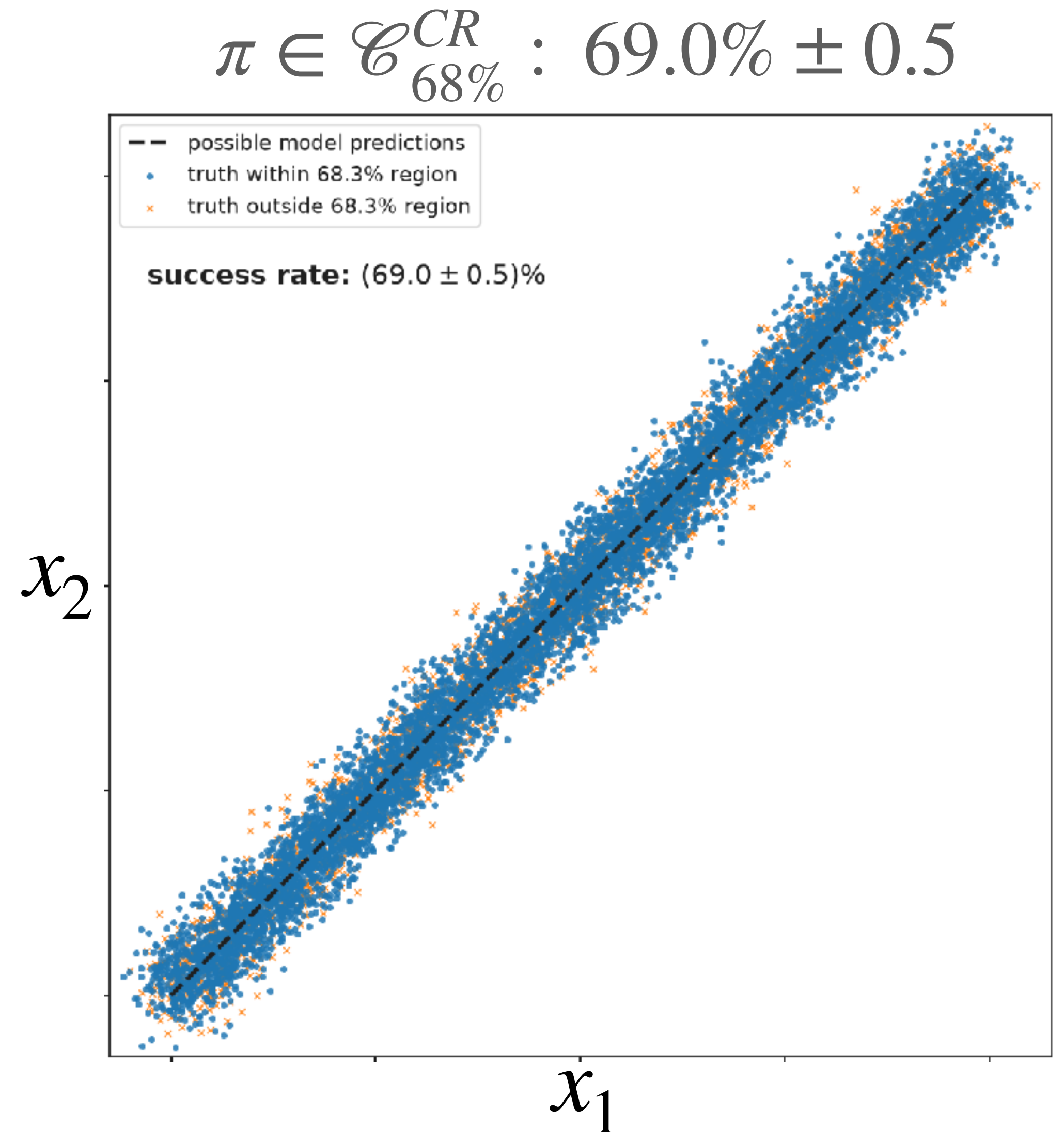


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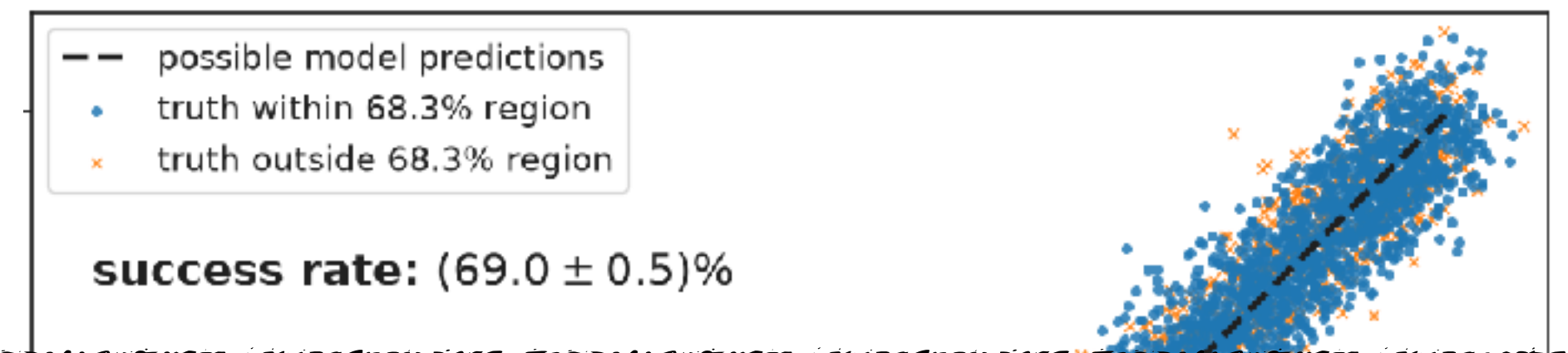
- Now repeat but with $\pi \sim p(\pi)$ each time.
- Calculate $\mathcal{C}_{68\%}$: is π_0 **inside** or **outside**?



Linear model $x[\pi]$

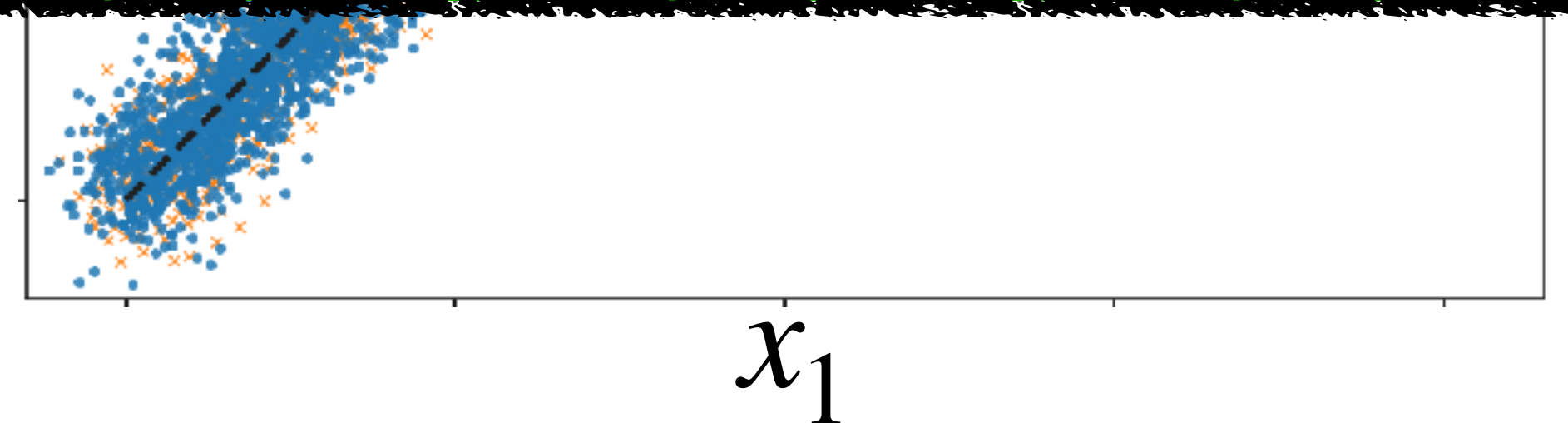
- This is our simple data generating model

$$\pi \in \mathcal{C}_{68\%}^{CR} : 69.0\% \pm 0.5$$



- Here, our Bayesian $\mathcal{C}_{68\%}^{CR}$ recovers the true π as frequently as $\mathcal{C}_{68\%}^{CI}$ does!
- Did you expect that?

- Calculate $\mathcal{C}_{68\%}$: is π_0 **inside** or **outside**?



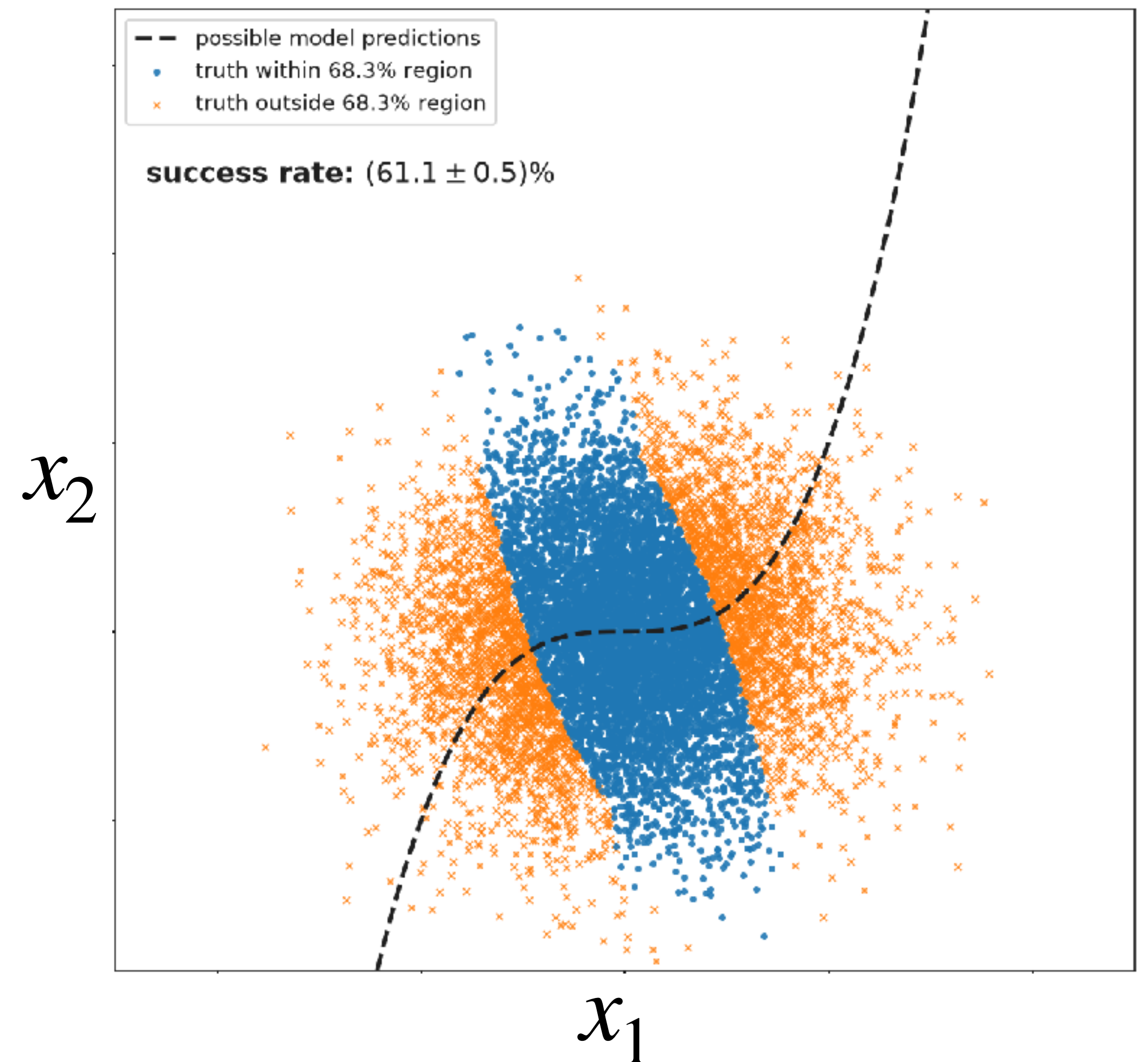
Non-linear model $x[\pi]$

- Now let's repeat but using:

$$\hat{x} \sim \mathcal{G} \left[\hat{x} \left| \begin{bmatrix} \pi \\ \pi^3 \end{bmatrix}, \begin{bmatrix} \sigma^2 & 0 \\ 0 & \sigma^2 \end{bmatrix} \right. \right]$$

- Repeat 10,000 experiments at $\pi = \pi^0$.
- Calculate $\mathcal{C}_{68\%}$: is π_0 **inside** or **outside**?

$$\pi \in \mathcal{C}_{68\%}^{CR} : 61.1\% \pm 0.5$$

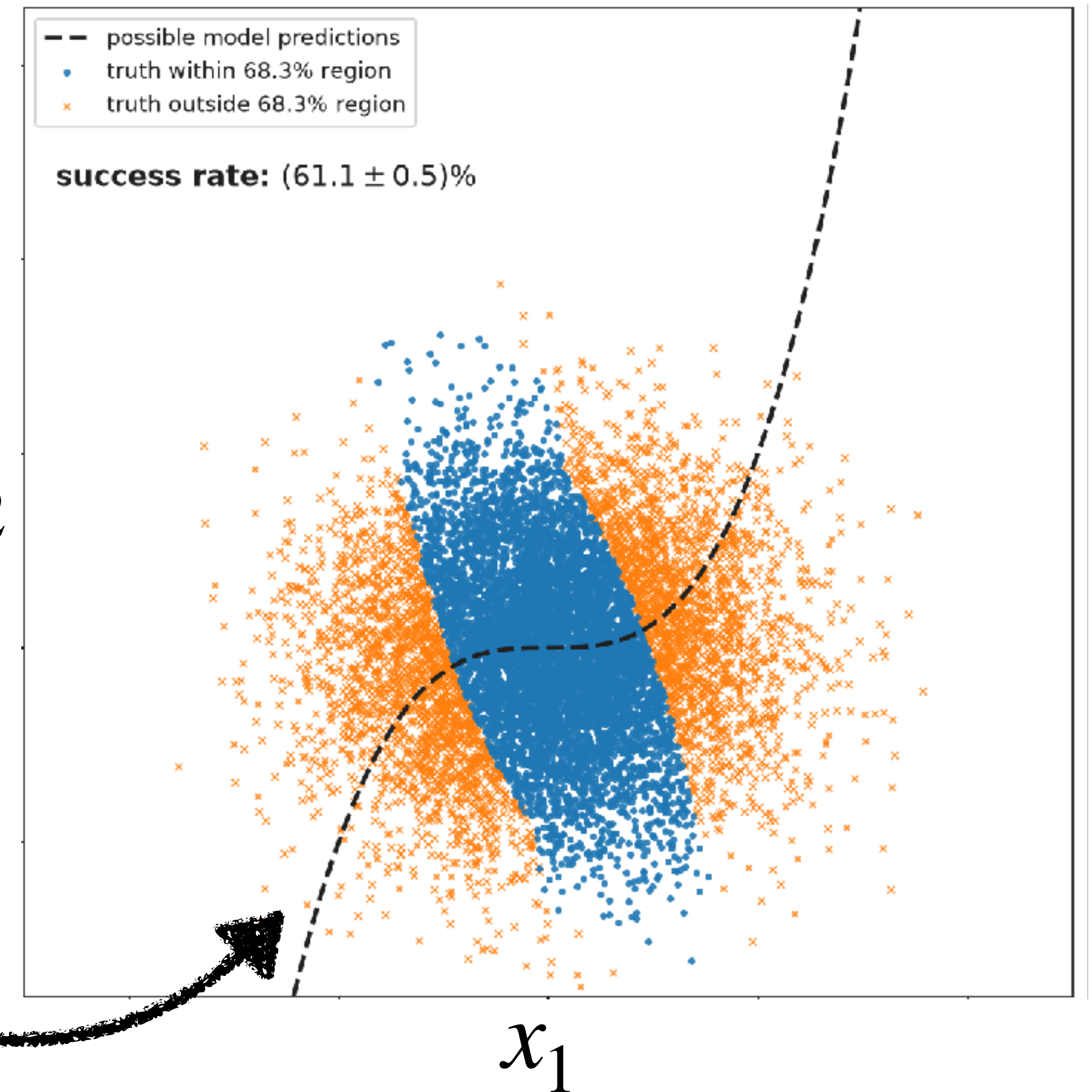


Non-linear model $x[\pi]$

- What causes the Bayesian $\mathcal{C}_{68\%}^{CR}$ to not recover π as expected?

$$p(\pi | \hat{x}) \propto \exp \left[\frac{1}{\sigma^2} \left\{ (\hat{x}_1 - \pi)^2 + \underline{(\hat{x}_2 - \pi^3)^2} \right\} \right]$$

$$\pi \in \mathcal{C}_{68\%}^{CR} : 61.1\% \pm 0.5$$



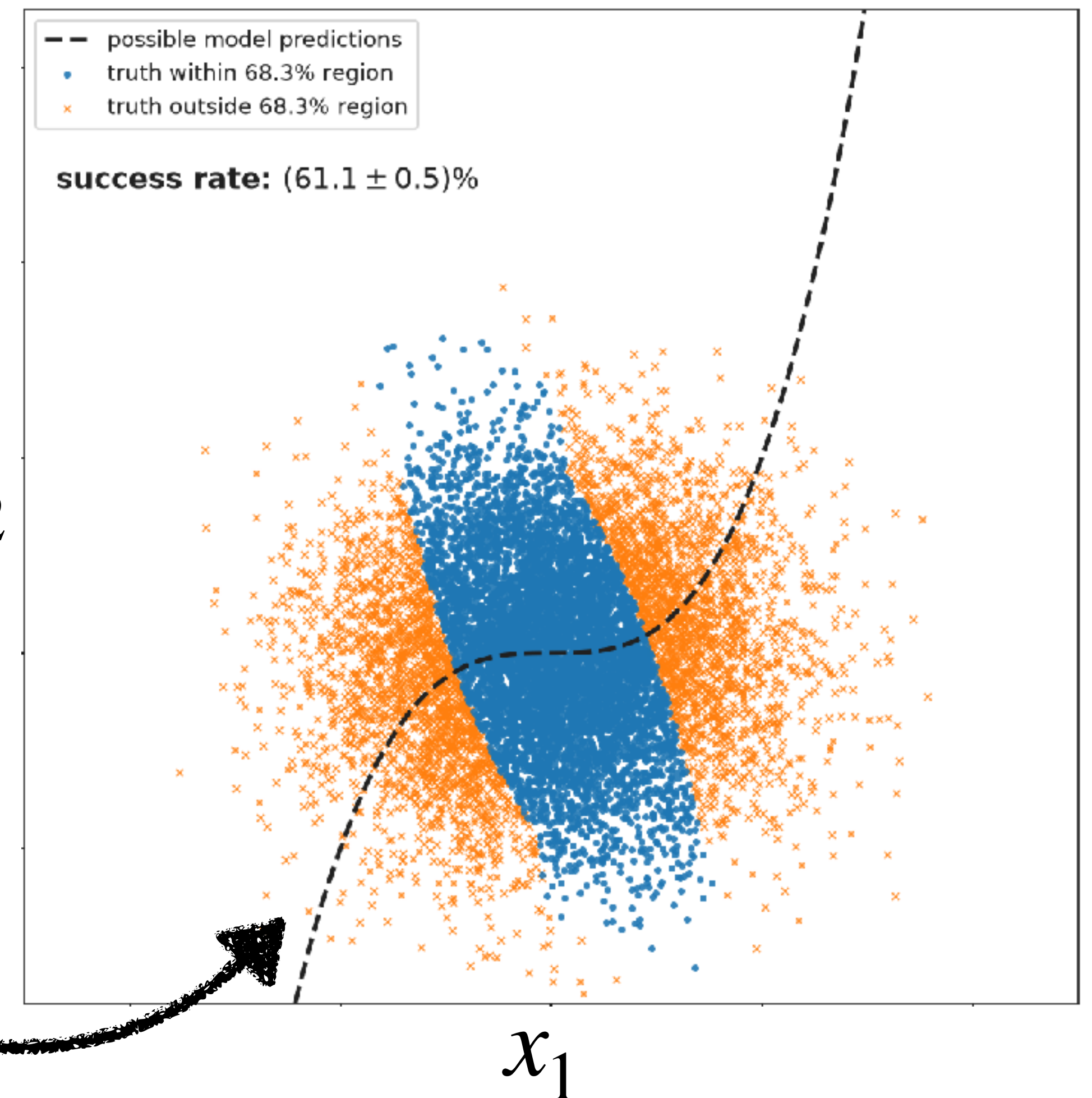
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- The maximum of $p(\pi | \hat{x})$ is where the $\mathcal{C}_{68\%}^{CR}$ starts,
 - but it is scattered since posterior is non-linear in π !

$$\pi \in \mathcal{C}_{68\%}^{CR} : 61.1\% \pm 0.5$$



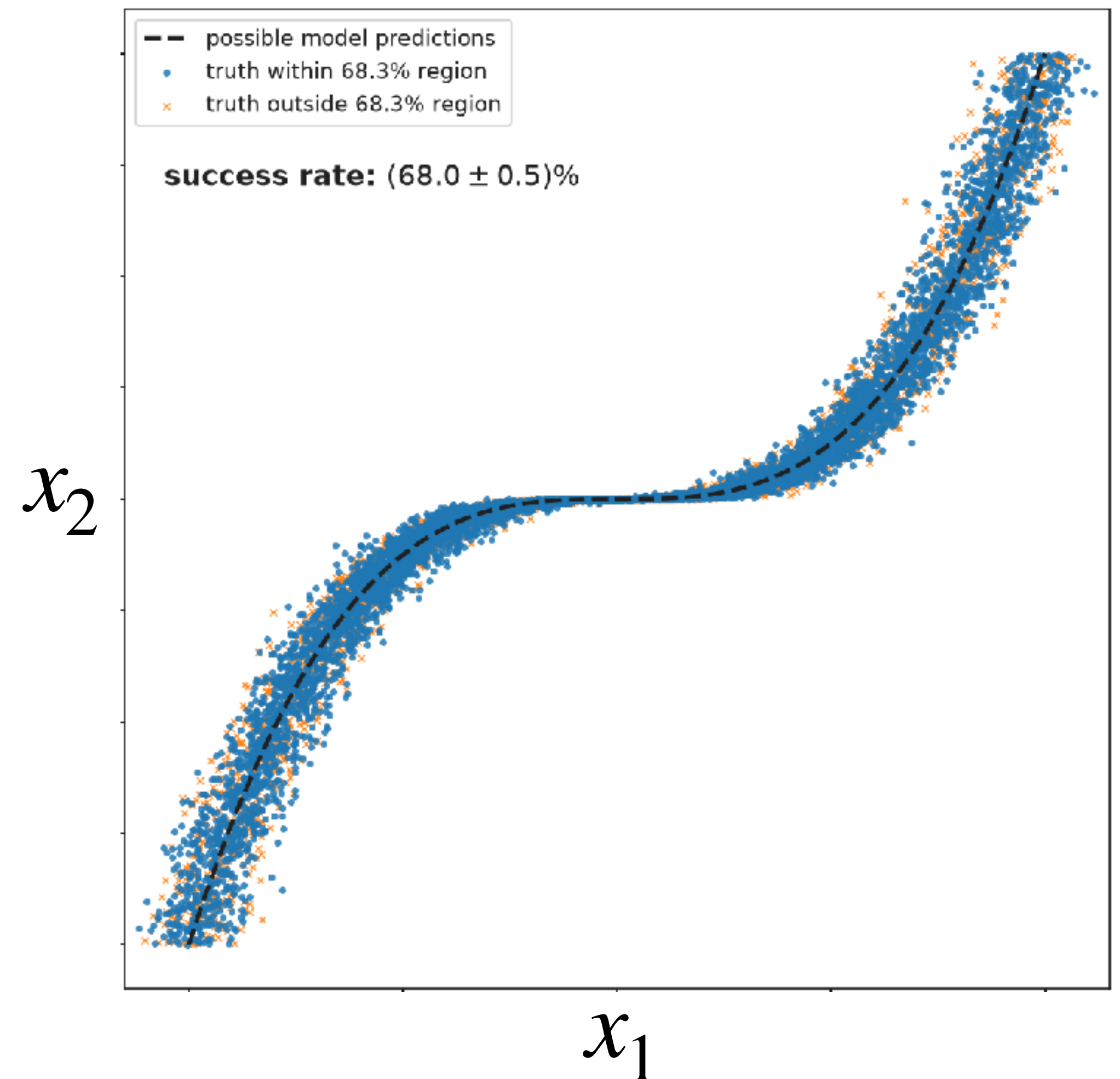
Non-linear model $x[\pi]$

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- Now repeat but with $\pi \sim p(\pi)$ each time.
- Calculate $\mathcal{C}_{68\%}$: is π_0 **inside** or **outside**?

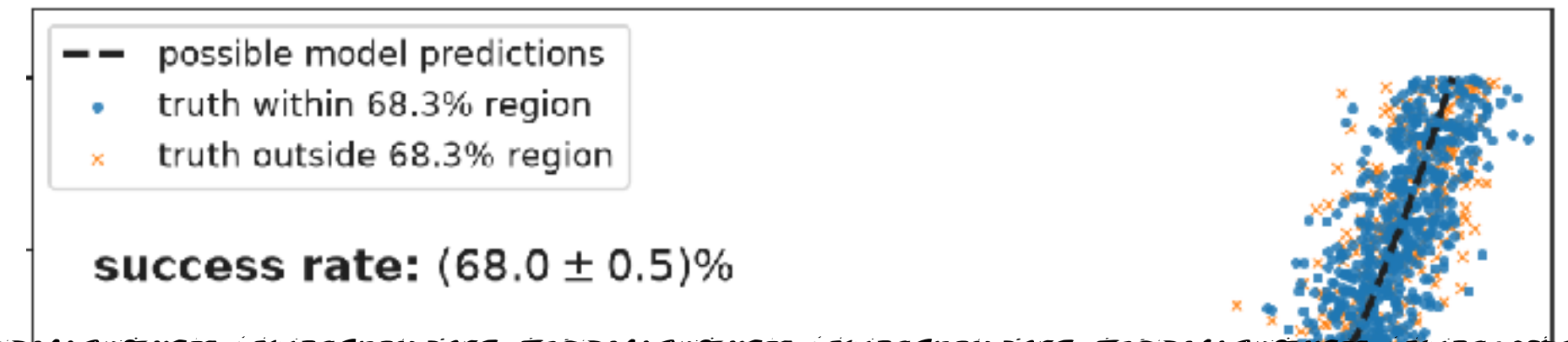
$$\pi \in \mathcal{C}_{68\%}^{CR} : 68.0\% \pm 0.5$$



Non-linear model $x[\pi]$

$$\pi \in \mathcal{C}_{68\%}^{CR} : 68.0\% \pm 0.5$$

- Now let's repeat but using:



- Here, our Bayesian $\mathcal{C}_{68\%}^{CR}$ does not recover the true π as frequently as

$\mathcal{C}_{68\%}^{CI}$ does!

- Did you expect that? Why?
- Calculate $\mathcal{C}_{68\%}$: is π_0 **inside** or **outside**?

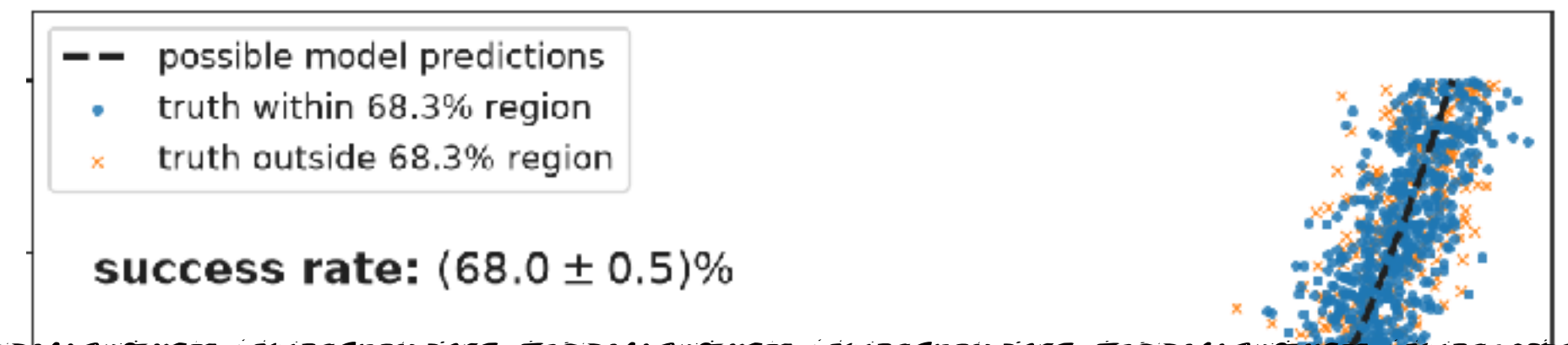


x_1

Non-linear model $x[\pi]$

$$\pi \in \mathcal{C}_{68\%}^{CR} : 68.0\% \pm 0.5$$

- Now let's repeat but using:



- Here, our Bayesian $\mathcal{C}_{68\%}^{CR}$ does not recover the true π as frequently as

$\mathcal{C}_{68\%}^{CI}$ does!

- Why did a non-linear model break the matching of $\mathcal{C}_{68\%}^{CI}$ and $\mathcal{C}_{68\%}^{CR}$?

- C

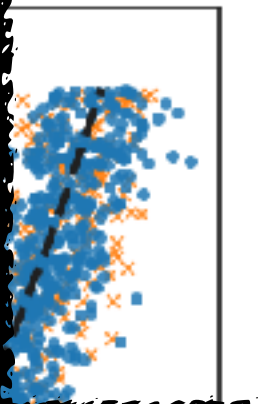
- $\hat{\pi}_{MAP}$ depends on the location of π ! Therefore so does $\mathcal{C}_{68\%}^{CR}$!

Non-linear model $x[\pi]$

$$\pi \sim \mathcal{D}^{CR} : 68.0\% \pm 0.5$$

Who cares?

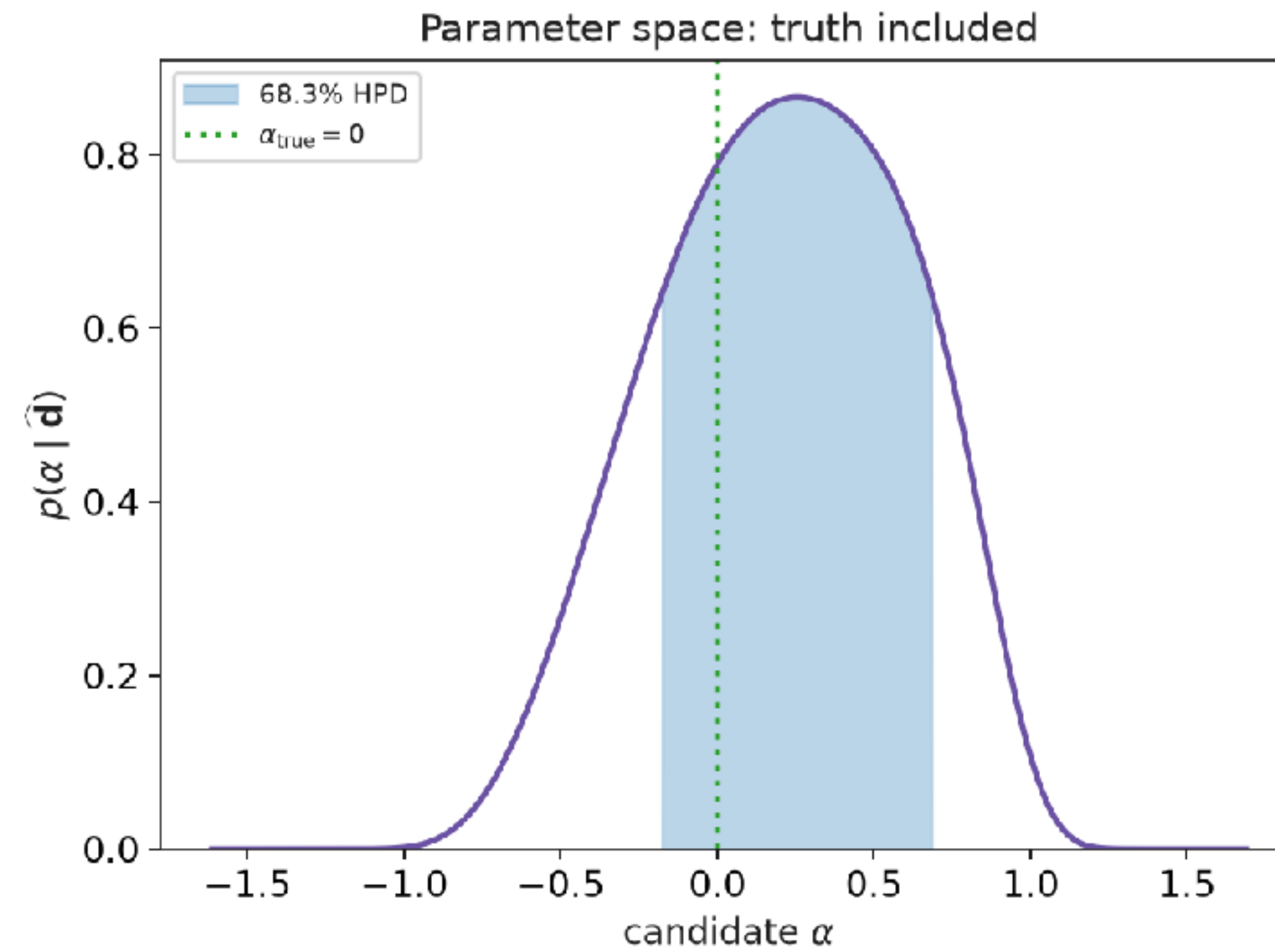
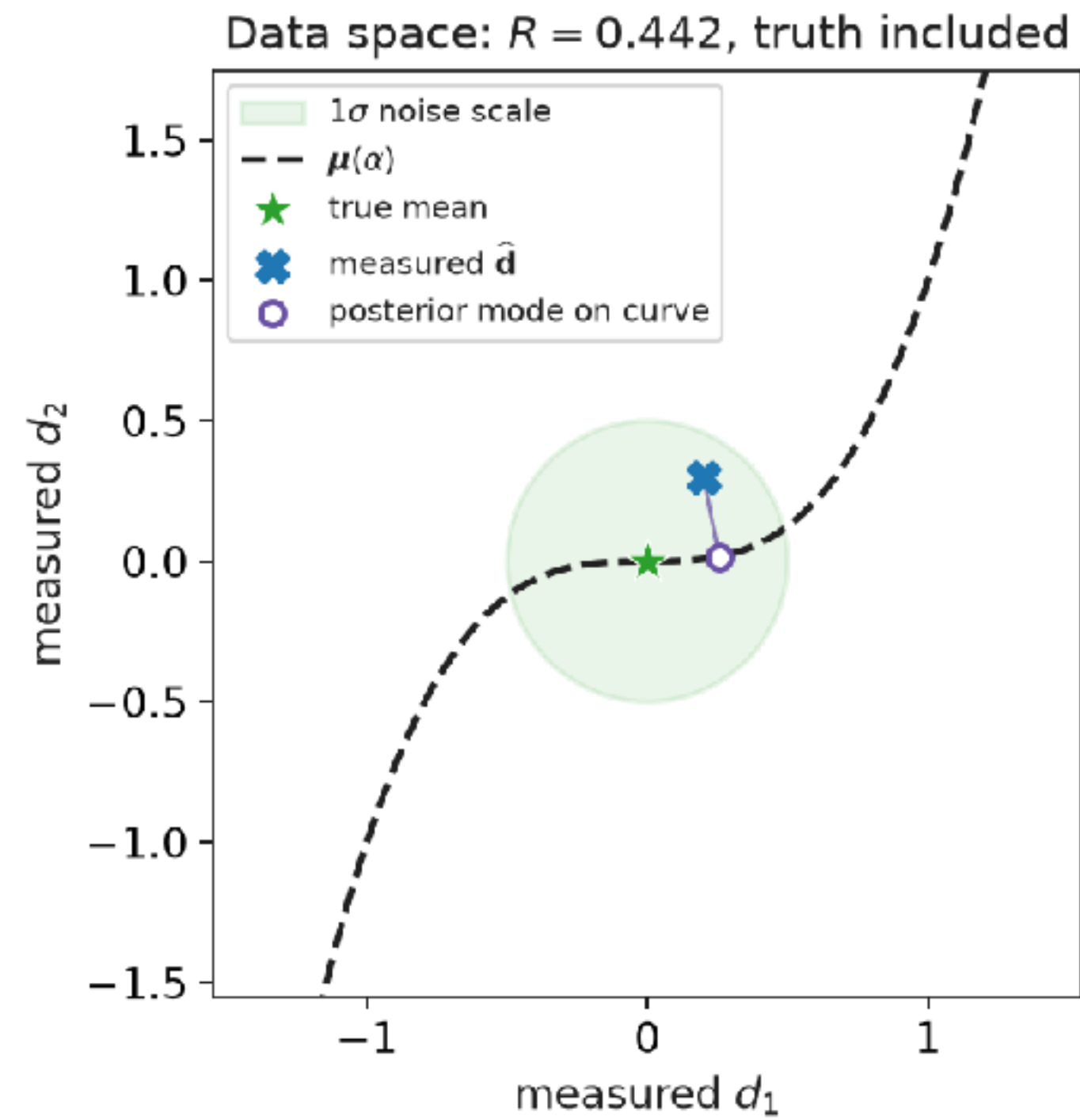
- Your Bayesian posterior does **not** recover the true parameters you are attempting to measure, as much as you may predict it does!



$\mathcal{C}_{68\%}^{CI}$ does!

-
- Why did a non-linear model break the matching of $\mathcal{C}_{68\%}^{CI}$ and $\mathcal{C}_{68\%}^{CR}$?
- $\hat{\pi}_{MAP}$ depends on the location of π ! Therefore so does $\mathcal{C}_{68\%}^{CR}$!

Non-linear model $x[\pi]$



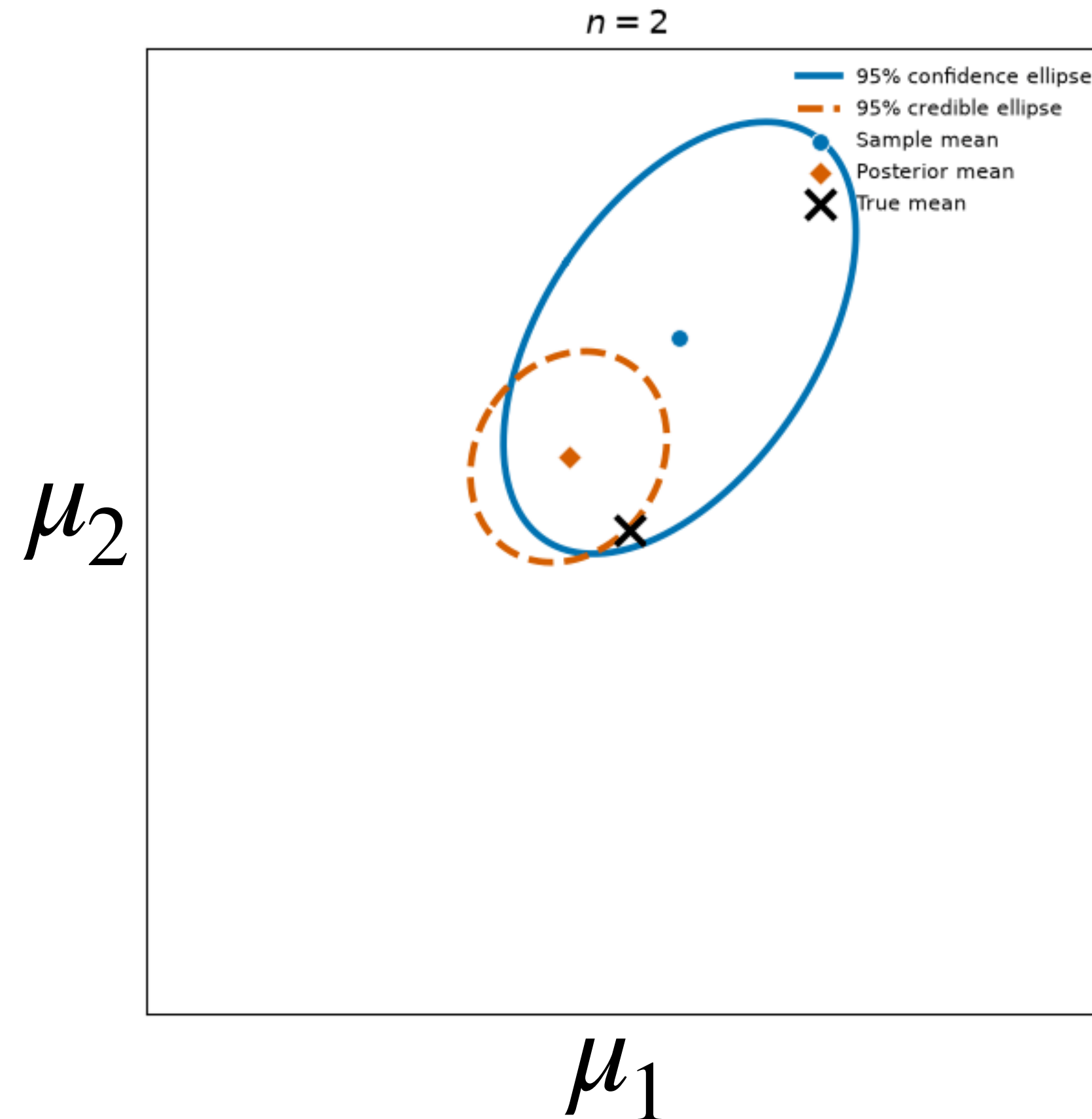
Bernstein von-Mises Theorem

- As n increases the endpoints of $\mathcal{C}_{\alpha\%}^{\text{CR}}$ and $\mathcal{C}_{\alpha\%}^{\text{CI}}$ agree.

$$p(\pi \mid \{x_i, \dots, x_N\}) \approx \mathcal{G}[\pi \mid \hat{\pi}_{\text{MLE}}, F_N[\pi]^{-1}]$$

- E.g. a 95% confidence interval contains 95% of the Bayesian posterior probability.

Bernstein von-Mises Theorem



$$p(\pi \mid \{x_i, \dots, x_N\}) \approx \mathcal{G}[\pi \mid \hat{\pi}_{\text{MLE}}, F_N[\pi]^{-1}]$$

Bernstein von-Mises Theorem

Who cares?

- A Bayesian posterior $p(\pi \mid \hat{\pi})$ is not guaranteed to recover the truth π in some fraction of repeated experiments.
- The BvM theorem dictates when it should; remember that Bayesians do not have to assume an objective truth π .

Write down something you learned today and
something you need to hear more about next lecture

 jed.homer@physik.lmu.de